



Contents lists available at ScienceDirect

Journal of International Economics

journal homepage: www.elsevier.com/locate/jie

Full length articles

Longevity and the value of trade relationships[☆]Ryan Monarch^{a,*}, Tim Schmidt-Eisenlohr^b^a Syracuse University, United States of America^b Federal Reserve Board, United States of America

ARTICLE INFO

Dataset link: <https://data.mendeley.com/datasets/yykpvjxsvj/1>

JEL classification:

F11

F14

L14

D22

Keywords:

International trade

Firm relationships

Learning

Trade dynamics

ABSTRACT

More than 80 percent of U.S. imports occur in preexisting firm-to-firm relationships, and disruptions to them can have large and long-lasting effects. Using U.S. Census data, this paper shows that as importers and their suppliers transact repeatedly, traded quantities and survival probabilities rise. We develop a general equilibrium trade model with relationship dynamics that is consistent with these facts. The quantification implies that long-term relationships are substantially more valuable to firms than new relationships, with wide variation across source countries. Losing relationship capital is costly for an economy, with disruptions to relationships causing notably lower levels of consumption and trade in the short- to medium term.

1. Introduction

Firm-to-firm relationships are the cornerstone of modern production, as intermediate inputs often pass between firms. International relationships may be destroyed when trade frictions rise and replacing them can be costly and time-consuming, creating long-term effects from supply chain disruptions. A large body of research has shown that relational contracts and learning about suppliers are central to firms' production decisions: considering shocks to those aspects of firm behavior has become especially relevant with the recent rise in global trade frictions and the disruptions caused by the COVID-19 pandemic. Exploiting U.S. Census data, this paper extends this line of research in two distinct ways. First, it documents stark differences between younger and older relationships: most relationships are new, while most trade takes place in existing relationships. Second, it embeds a model featuring

[☆] We thank George Alessandria, Bernardo Blum, Fabrice Defever, Brian Green, Brian Kovak, Florian Mayneris, Andreas Moxnes, Ameet Morjaria, Emanuel Ornelas, James Rauch, Peter Schott, and Olga Timoshenko for helpful discussions, as well as seminar participants at the Federal Reserve Board, the World Bank, the Rocky Mountain Empirical Trade conference, the Society for Economic Dynamics conference, the Washington Area International Trade Symposium, the European Bank for Reconstruction and Development, American University, the Society for Institutional & Organizational Economics, USITC, Nottingham University, Stockholm University, Uppsala University, UQAM, University of Montreal, Carnegie Mellon University, University of Maryland, University of Virginia, the Southern Economic Association meetings, the Stanford RDC Conference, the European Trade Study Group, the Federal Reserve Board International Trade Conference, and the Empirical Investigations in International Trade conference for helpful comments. This work was undertaken while Monarch and Schmidt-Eisenlohr were under Special Sworn Status at the U.S. Census Bureau. Any opinions and conclusions expressed herein are those of the authors and do not necessarily represent the views of the Federal Reserve Board, or anyone affiliated with the Federal Reserve System. Any views expressed are those of the authors and not those of the U.S. Census Bureau. The Census Bureau's Disclosure Review Board and Disclosure Avoidance Officers have reviewed this information product for unauthorized disclosure of confidential information and have approved the disclosure avoidance practices applied to this release. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2407. (CBDRB-FY22-P2407-R9906, CBDRB-FY24-P2407-R10909).

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<https://doi.org/10.1016/j.jinteco.2023.103842>

Received 29 April 2021; Received in revised form 5 November 2023; Accepted 6 November 2023

Available online 8 November 2023

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relationship dynamics into a general equilibrium trade framework. Calibrating the model to the data and solving for the dynamic evolution of trade patterns, we quantify the value of relationships to firms and estimate the long-run impact that relationship disruptions have on trade flows.

We start by establishing several facts. First, although in a given year, the majority of relationships between U.S. importers and foreign suppliers are newly formed, more than sixty percent of U.S. imports by value take place within relationships that are 3 years or older. Such a breakdown is masked in more aggregated data, where trade is highly concentrated in experienced firms and continuously imported products. Second, the survival probability of a relationship increases with age and is affected differently by different aggregate shocks: During the Global Financial Crisis, new relationships were less likely to survive while older relationships were more likely to survive, whereas during the U.S.–China trade war, survival declined across all relationship ages. Third, over the course of a relationship, traded amounts increase while prices decline. Finally, there is substantial heterogeneity across source countries in these patterns.

Motivated by these findings, we present a model of importer–exporter relationship dynamics. In line with the empirical results, the model features a connection between the survival probability of a relationship, the amount traded within that relationship, and the age of the relationship.¹ The value of a relationship to a firm is defined as the expected sum of profits from that relationship relative to the expected sum of profits from a new relationship, an expression that varies both by source and by relationship age.

The setup is then embedded in a two-country general equilibrium trade model that builds on [Krugman \(1980\)](#) and is calibrated to U.S. firm-to-firm import data. We estimate separate parameters for each of 10 major trading partners of the United States and for the World as a whole, using survival rates and growth rates in trade volume across age cohorts as target moments. The model generates a good fit to the targeted moments, and also matches well the cross-sectional distributions of relationships and trade that are not directly targeted. Computing relationship values with this calibration implies that long-term relationships are, on average, 21 times more valuable to firms than new relationships, with a maximum of 72 (Korea) and a minimum of 5 (France).

Our model provides a natural way to consider the longer-run effects of policies that are disruptive to trade relationships. We conduct two experiments. First, we study what happens when there is a one-off symmetric shock to relationship survival. Specifically, we assume that 5 percent of relationships in each age group are destroyed on impact, with entry also being subdued in the shock period. After the initial shock, entry rises notably, and the share of young relationships is above its steady state-level over the transition path. As younger relationships are less efficient, trade and consumption fall below their steady-state values and only slowly transition back to their steady-state levels as relationship-capital is rebuilt. Imports from source countries with younger relationships on average, such as China, fall by less and recover more quickly, as older relationships which take longer to rebuild are less important for these sources.

Second, we study a shock that matches the changes in survival rates across age cohorts during the U.S.-China trade war. Specifically, we shock the survival rates of U.S.-China relationships in the initial period to exactly match their exact decline in 2018. We then solve for the transition back to the original steady state. Again, this shock reduces efficiency and lowers trade and consumption. However, the U.S.-China trade war disproportionately lowered survival of younger relationships, who are less likely to survive to begin with. Hence, the declines in the delivery probability, trade, and consumption are less pronounced compared to a symmetric shock of the same size that hits all ages equally. For the same reason, the trade war shock also features a faster recovery than a symmetric shock.

Our paper has three distinct contributions. First, it documents the importance of trade relationships, their differences across countries, and how they respond to crisis episodes, using the universe of U.S. import data. Second, it embeds a reduced-form model of relationship dynamics into a trade model that, after calibrating to the data, allows us to calculate the value of trade relationships to importers. Third, solving for the dynamic transition path of the economy, we evaluate the effects of relationship disruptions on the economy in the short- to medium-term.

There is a burgeoning literature that, like our paper, uses “two sided” international trade data to study relationships between firms. [Blum et al. \(2013\)](#) use linked data on importers and exporters in Latin America, showing that firms that occasionally export often export the same goods to the same importers. [Eaton et al. \(2014\)](#) study relationships between Colombian exporters and U.S. importers. They calibrate a search and matching model with learning to match exporter decisions, including sales, number of clients, and transition probabilities. [Eaton et al. \(2016a\)](#) develop a model of relationship formation based on searching by both buyers and sellers. Firms are heterogeneous in their cost of searching for effective partners, and relationships end for exogenous reasons. [Kamal and Sundaram \(2016\)](#) use the same data as our paper to determine how likely Bangladeshi textile producers are to follow other exporters in the same city in exporting to a particular partner. [Monarch \(2022\)](#) applies a dynamic discrete choice model to the sourcing decisions of U.S. importers over time to estimate the cost of switching suppliers. Two-sided trade data are also used to study the effects of firm heterogeneity on trade: [Bernard et al. \(2018\)](#) use Norwegian trade data to develop a model of relationship-specific fixed costs of exporting.² [Carballo et al. \(2018\)](#) look at relationships in several Latin American countries and develop a model to analyze the role of competition. [Heise \(2015\)](#) studies the effects of firm-to-firm relationships on price rigidity and exchange-rate pass-through using U.S. importer data and finds that prices grow within a relationship as trade increases. We complement this

¹ The success probability of a relationship can be micro-founded by a learning mechanism, as we show in Section 3.4. There are several alternatives to a learning model that one could consider. While simple models of long-term contracts or relationship switching costs would be inconsistent with our finding that trade increases within relationships over time, a model of relationship-specific investment could generate similar patterns. However, to match the data, such a model would have to feature relationship dynamics that differ across countries.

² [Benguria \(2021\)](#) also studies two-sided heterogeneity, using data from Colombia.

literature by focusing on cross-country heterogeneity in relationship patterns and using it to quantify the value and the dynamic effects of relationships.³

A set of papers directly study the role of long-term relationships for international trade flows.⁴ Egan and Mody (1992) provide survey evidence that importers from developed countries initiate emerging-economy trade relationships with very small purchases. Rauch and Watson (2003) rationalize this finding in a model with importer learning. In their model, relationship persistence is the result of a dynamic trade-off between the per-period costs of a supplier and the reliability of other potential partners. McLaren (1999) examines the circumstances under which firms base their relationships on trust instead of contracts, and uses it to understand international differences in supplier relations. Macchiavello and Morjaria (2015) study Kenyan rose exporters and find that the value of a relationship increases with age. They also show that in long-run relationships, buyers have already learned the type of the seller, and, therefore, no costly signaling is necessary in times of crisis. Heise et al. (2015) study how changes in trade policy may affect the organization of supply chains.

One way to understand the facts we present is through a model of learning over the course of a trade relationship. The learning model we describe in Section 3.4 as a plausible micro-foundation for our analysis directly builds on Araujo et al. (2016), who study learning by exporters and test reduced-form predictions with Belgian firm-level data.⁵ In the tradition of the seminal work of Jovanovic (1982), a number of other papers consider firm learning in different contexts, such as its relationship with exports (Albornoz et al., 2012), foreign demand (Ruhl and Willis, 2017), growth (Arkolakis et al., 2018), and international prices (Bastos et al., 2018).

Finally, our work adds to the literature on trade dynamics. Alvarez and Lucas (2007) develop a dynamic solution to the general equilibrium trade model of Eaton and Kortum (2002), while Eaton et al. (2016b) embed this framework in an international real business cycle model. Several papers study how firm behavior shapes aggregate trade dynamics. Ruhl (2008) examines the extensive margin of trade and its implications for trade elasticities. Atkeson and Burstein (2010) model the dynamic product and process innovation decision of exporters. Alessandria and Choi (2014) show that long-term gains of tariff cuts are larger when not only sunk costs of exporting, but also continuation costs, are taken into account. Alessandria et al. (2018) consider the sluggish response of trade volumes to changes in trade barriers or price shifts by modeling the buildup of a firm's exporting capacity over time.

The rest of the paper is organized as follows. Section 2 describes the importer–supplier data and presents empirical findings on trade relationships. Section 3 discusses the model of relationship dynamics and how it fits into a general equilibrium framework. Section 3 also introduces a potential microfoundation for the model. Section 4 describes the calibration exercise and the quantitative evaluations of the value of trade relationships both to firms and to the economy. Section 5 concludes.

2. Empirics

2.1. Data definitions and cleaning

The data come from the Longitudinal Foreign Trade and Transaction Database (LFTTD), which is collected by U.S. Customs and Border Protection and maintained by the U.S. Census Bureau. Every transaction in which a U.S. company imports or exports a product requires the filing of Form 7501 with U.S. Customs and Border Protection, and the LFTTD contains the information from these forms.⁶ In this paper, we utilize the import data, which include quantity and value exchanged for each transaction, Harmonized System (HS) 10 product classification, date of import and export, port information, country of origin, and a code identifying the foreign exporting partner.⁷ Known as the *manufacturing ID*, or *MID*, the foreign partner identifier contains limited information on the name, address, and city of the foreign supplier.⁸ Kamal and Monarch (2018) find substantial support for the validity of the MID and its applicability towards studying buyer–seller relationships, and Redding and Weinstein (2017) show that many of the salient features associated with exporting activity (such as the prevalence of multi-product firms and high rates of product and firm turnover) are replicated for MID-identified exporters.⁹ Pierce and Schott (2016), Monarch (2022), Kamal and Sundaram (2016), Eaton et al. (2014), Heise (2015) and Handley et al. (2023) have all used this variable in the context of studying U.S. firm relationships in international trade.

³ More recently, Benguria et al. (2023) show that long-term relationships can also be beneficial financially: by making it easier to use trade credit, they lower borrowing costs and thereby improve firms' finances.

⁴ Earlier work on buyer–supplier relationships in international trade centered on the study of networks. Rauch (2001) surveys the potential for transnational cultural networks to reduce barriers to entry, and Rauch and Watson (2004) present a model where economic agents use their networks to produce or export more efficiently or to become an intermediary. Krauthaim (2012) models how information sharing in networks may affect the fixed costs of exporting and thereby the measured effects of distance on trade. Chaney (2014) studies a dynamic model where firms search for additional trade opportunities through their networks.

⁵ Our work is also related to the wider literature on learning and entry. Impullitti et al. (2013) develop a model of entry and exit where firms pay sunk costs to start exporting and face persistent productivity shocks. Besedeš and Prusa (2006) study the effect of product differentiation on U.S. importer relationships with their suppliers. Besedeš (2008) finds that reliable suppliers lead to longer relationships and larger export orders, while only a small fraction of relationships end as a result of switching behavior. Timoshenko (2015a) uses Colombian firm-level data to distinguish between the role of sunk costs and learning for the persistence of exporting. Timoshenko (2015b) finds that new exporters do more product switching, and develops a model of learning that rationalizes this finding.

⁶ Approximately 80 to 85 percent of these customs forms are filled out electronically (Krizan, 2012).

⁷ Importantly, only the import data contains information on both sides of a trade transaction; the U.S. export data does not identify the purchaser of the exported goods.

⁸ Specifically, the MID contains the first three letters of the producer's city, six characters from the producer's name, up to four numeric characters from its address, and the ISO2 code for the country of origin.

⁹ Kamal and Monarch (2018) describe the institutional background for the creation of the variable, propose various cleaning methods, and compare the number and composition of exporters from U.S. data with international data.

Table 1
Relationship structure of U.S. imports, 2011.

Panel A: Importer–supplier			
	New	1–2 years	3 or more years
Relationship share	62.0	24.1	13.9
Trade share	13.4	22.5	64.1
Panel B: Importer–source			
	New	1–2 years	3 or more years
Relationship share	43.7	23.1	33.2
Trade share	3.4	4.8	91.8
Panel C: Product–source			
	New	1–2 years	3 or more years
Relationship share	25.2	15.6	59.2
Trade share	2.8	2.7	94.6

Note: A new relationship is one that is not found in any previous year of data. In Panel A, a relationship is defined as a U.S. importer transacting with a foreign supplier. In Panel B, a relationship is defined as a U.S. importer transacting with a foreign source country. In Panel C, a relationship is defined as U.S. imports of a product from a foreign source country.

For our analysis, we eliminate “related-party” transactions, as well as U.S. importers whose domestic operations are classified as wholesale or retail. We follow the methods of [Bernard et al. \(2009\)](#) for cleaning the LFTTD. Specifically, we drop all transactions with imputed quantities or values (which are typically very low-value transactions) or converted quantities or values.

Finally, some definitions: an **importer** is a U.S. importing firm, while a **supplier** is a non-U.S. entity identified by the MID as exporting to the United States. A **relationship** is an observation of an importer–supplier combination. The **age** of a relationship is how many *consecutive* years that relationship has appeared in the U.S. import data, with the first observation of a relationship considered to be age 0. In other words, a relationship of age 1 is one that has appeared in two consecutive years of data, and so on. We call the distribution of relationship counts over different ages the **count distribution**, and the distribution of trade values across relationships of different ages the **value distribution**. Data from both distributions are represented as shares of the total. Finally, **source** refers to the country in which the supplier is located.

2.2. Empirical findings

2.2.1. Count and value distributions of U.S. imports

[Table 1](#) Panel A presents a breakdown of U.S. arm’s-length imports in 2011 by the age of relationships. Two points stand out. First, the count distribution is skewed to the right. New relationships account for the clear majority (62%) of total relationships in 2011, while only about 14% of relationships are age 3 or older. Second, the value distribution is skewed to the left. Long-term relationships – those age 3 or older – account for the clear majority (64%) of total imports in 2011. Only about 13% of imports occur in new relationships.¹⁰

The finding that most relationships are new, while most trade is occurring in old relationships is robust to a number of different cuts of the data. First, throughout the paper, as is common in empirical studies of firms participating in international trade, we exclude wholesale and retail firms from the analysis.¹¹ However, as shown in Panel A of [Table B.1](#), the key findings – the right skew of the count distribution and the left skew of the value distribution – are qualitatively unchanged from our baseline when including wholesale and retail firms. The findings are also robust to a different definition of age. In particular, we define a relationship’s “broad age” to be the difference between the current year and the first year a relationship appeared in the data, rather than the number of consecutive years. For example, this would classify a relationship with 2 observations 5 years apart as a relationship of “broad age” 5, while also classifying a relationship with 5 observations over those 5 years as “broad age” 5. This more liberal definition of relationship age mechanically increases the share of long-term relationships, as shown Panel B of [Table B.1](#). However, with this alternative definition, it continues to be the case that most relationships are new and that the bulk of imports occur in long-term relationships.¹²

In contrast, employing relationship-level information is key to our main findings. Indeed, the distributions reported in [Table 1](#) change substantially when they are calculated from more aggregated data. Panel B of [Table 1](#) defines a relationship as an importer–source pair, rather than an importer–supplier pair. In this case, older “relationships” appear more than 2 times more prevalent compared to Panel A (33% versus 14%). Furthermore, the share of trade in younger relationships (new, age 1, or age 2) drops from

¹⁰ We do not study related-party trade relationships in this paper. These relationships are systematically different from arm’s length relationships and might therefore exhibit very different dynamics. They are also likely to be longer lasting. [Kamal and Monarch \(2018\)](#) demonstrate that related-party relationships are also more concentrated, accounting for 40% of imports but only 7% of all relationships.

¹¹ See, for example, [Hillberry and Hummels \(2003\)](#) and [Bernard et al. \(2010\)](#).

¹² In line with the analysis in [Fanelli and Hallak \(2018\)](#), [Table B.2](#) illustrates how common it is for relationships with a “broad age” to have gaps. For relationships of broad age 5, over half of relationships, and close to 90% of imports, have no interruption.

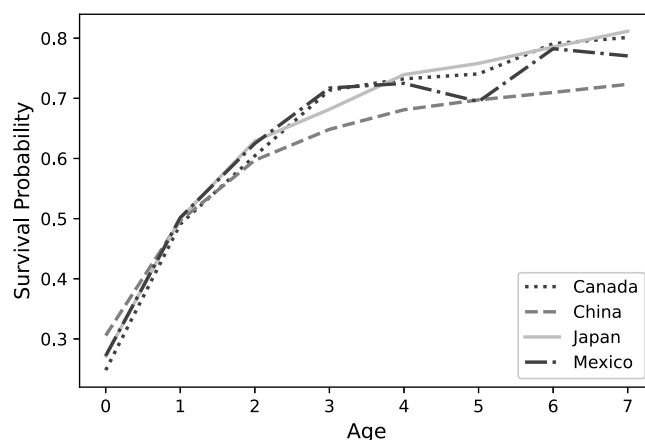


Fig. 1. Relationship survival probabilities. Note: This figure plots the conditional survival probability by age for relationships between U.S. importers and foreign suppliers from each of four major import sources of the United States in 2011; that is, the share of relationships of age k that survive to age $k + 1$.

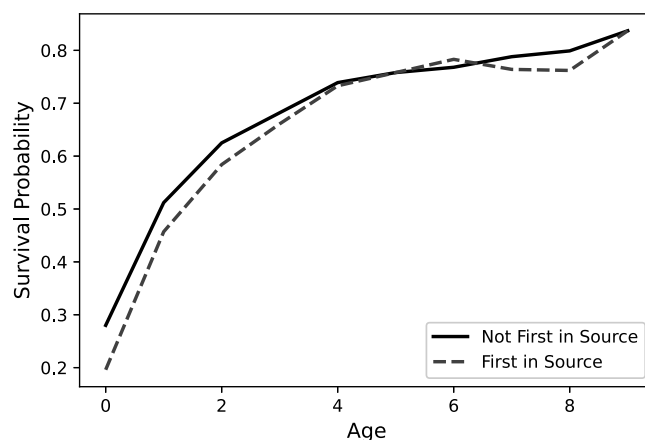


Fig. 2. Relationship survival probability, experienced vs. new importers. Note: This figure plots the conditional survival probability by age for relationships between U.S. importers and all foreign suppliers in 2011; that is, the share of relationships of age k that survive to age $k + 1$. The solid line is the survival probability for relationships that are not the first relationship for a U.S. importer with a particular source, while the dashed line is the survival probability for relationships that are the first relationship for a U.S. importer with a particular source.

35% in Panel A to less than 10%, demonstrating a much more pronounced skew to the value distribution. Thus even relying on importer-source level data – already a relatively disaggregated measure – masks the relative importance of younger relationships in *both* dimensions. Zooming out even further, Panel C of Table 1 shows that the relevance of younger trade relationships is even less apparent in aggregate U.S. import data. Defining a relationship at the HS10 product-source level, both the count distribution and the value distribution are even more skewed towards older relationships. Indeed, although 35% of all U.S. imports are in importer–supplier relationships less than 3 years old, only about 6% of U.S. imports are in product-source “relationships” less than 3 years old. In this sense, Panel B (importer-source) and C (product-source) are more similar to each other than they are to the relationship level data, indicating the importance of considering trade at the importer–supplier level.

2.2.2. Relationship survival

The survival probability increases as relationships age. We next explore the survival probability of a trade relationship. Fig. 1 shows the conditional survival probability of a relationship by age for the top 4 import sources of the United States in 2011: what share of relationships of age k in 2010 survived in 2011? The figure shows that the older a relationship is, the higher the likelihood that relationship survives an additional year. About 30% of new relationships survive, compared to 60% of age 4 relationships and 80% of age 7 relationships.¹³ The figure also displays notable cross-source heterogeneity. For example, the survival of relationships with Chinese suppliers is lower for almost all ages compared to the survival of relationships with Japanese suppliers.

¹³ Inspired by Fanelli and Hallak (2018) and Besedeš and Prusa (2006), we also examine how common re-entry of exiting relationships is. Of all relationships started in 2010 that exited in 2011, only 4.8% re-entered in 2012.

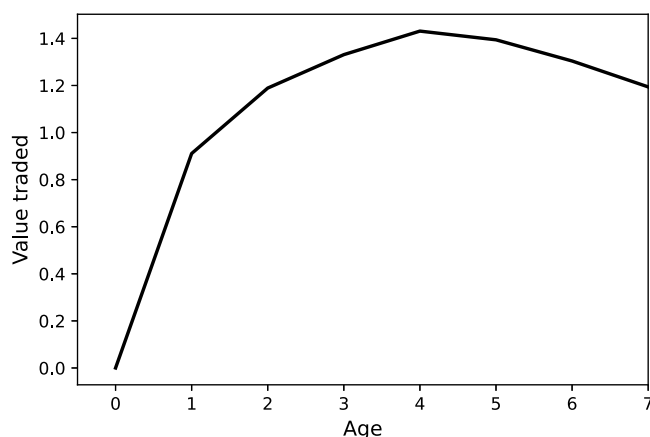


Fig. 3. Evolution of trade within a relationship. Note: This figure plots the coefficients for ages 1 through 7 from a regression of log imported value on relationship age dummies for relationships that were 10 years old in 2011 as in Eq. (1). The regression includes relationship fixed effects and importer-year fixed effects. The regression uses data from 2001 through 2011.

It is again important to consider whether this increase in survival over the course of a relationship is primarily driven by importer-level factors rather than relationship-level factors. For example, if older relationships mostly occur for older importers, then it would be more appropriate to analyze these trends at the importer level instead of the relationship level. Fig. 2 contrasts the survival probability of a relationship that is the first relationship for an importing firm in that source (dashed line) with the survival probability of a relationship that is not the first relationship for an importing firm (solid line). Interestingly, the boost in survival over time is just as large for new relationships in “familiar” sources to an importer, indicating that relationships are the key dimension along which this improvement occurs. In other words, the survival probability for relationships increases with age, even if the U.S. importer has imported from that source before. There is within-relationship improvement above and beyond any importer-source specific factors.

The importance of relationship-specific traits in explaining survival can also be seen by examining the factors that cause relationships to end. Following Alessandria et al. (2021), we define an indicator variable for whether a relationship between importer m and supplier x stopped transacting in a given year t ($Stop_{mxt}$). We then examine the factors that are correlated with that stoppage, including the import value in the relationship, the import value for that importer, and whether year t is the first/second year of that relationship or import source for importer m . Column 1 of Table 2 shows that relationships with less history (meaning the relationship either started that year or the previous year) are more likely to end, while Column 2 shows that relationships in which the importer has less source experience are also more likely to end. Including both of these factors together allows us to disentangle the effect of source experience from the effect of relationship experience. Column 3 shows that even controlling for importer-source experience, relationship inexperience affects relationship death. In fact, the sizes of the coefficients indicate that relationship inexperience is more important for predicting relationship death than source inexperience.

Relationship survival changes during crisis episodes. What happens to relationships during crisis episodes? Are older relationships more likely to survive adverse shocks? To answer this question, we look at relationship survival around the Global Financial Crisis in 2007/2008 and around the U.S.-China trade war in 2018/2019. In particular, we ask how relationship survival changed across the relationship age distribution in crisis times compared to the preceding years.

First, we look at relationship survival around the Global Financial Crisis. In particular, we compare the average probability over the years 2004–2006 for U.S. importer-foreign supplier relationships to survive an additional year to the same probability in 2007. As shown in Table 3, during the Global Financial Crisis, the survival probability of older relationships increased compared to pre-crisis years. In contrast, the survival probability of new relationships declined by 4 percentage points during the crisis.

Second, we compare the average survival probability for U.S.-China trade relationships over the years 2014–2016 to the same survival probability in 2017, in which survival into 2018 was affected by the massive increase in U.S. tariffs on Chinese imports in 2018. Interestingly, as shown in Table 4, changes to relationship survival are quite different from those during the Global Financial Crisis. For the U.S.-China trade war shock, the data shows that survival declined for all relationship ages, with the biggest percentage declines for new relationships. While we leave studying the ultimate reasons why relationships respond differently to these two crisis episodes to future research, we use these observed exit rates in Section 4.3 as inputs into our counterfactual analysis.¹⁴

¹⁴ Handley et al. (2023) show that the U.S. import decline during the U.S.-China trade war was driven primarily by relationships ending, fewer relationships forming, and U.S. firms exiting from sourcing in foreign markets.

Table 2
Stopper regressions.

	$Stop_{mxt}$	$Stop_{mxt}$	$Stop_{mxt}$
$Value_{mt}$	−0.005*** (0.001)	−0.001 (0.001)	−0.004*** (0.001)
$Value_{mxt}$	−0.065*** (0.002)	−0.082*** (0.002)	−0.065*** (0.002)
$FirstYear_{mxt}$	0.257*** (0.011)		0.244*** (0.011)
$SecondYear_{mxt}$	0.095*** (0.004)		0.088*** (0.004)
$SourceFirstYear_{mxt}$		0.147*** (0.008)	0.039*** (0.006)
$SourceSecondYear_{mxt}$		0.076*** (0.003)	0.032*** (0.002)
N	1,372,000	1,372,000	1,372,000
FE	$t, HS2$	$t, HS2$	$t, HS2$

Note: This table presents the results of a regression of an indicator variable for whether a relationship between importer m and supplier x stopped in a certain year t ($Stop_{mxt}$) on factors that may be correlated with that stoppage. The relationship stopped if it is not observed the following year. $Value_{mt}$ represents the log import value for importer m in year t , while $Value_{mxt}$ is the log import value for the relationship between m and x in year t . $FirstYear_{mxt}/SecondYear_{mxt}$ are indicator variables for whether t is the first/second year that m and x have transacted, while $SourceFirstYear_{mxt}/SourceSecondYear_{mxt}$ are indicator variables for whether t is the first/second year that m transacted with the source country c of supplier x . The regression includes year fixed effects as well as fixed effects for the most imported HS2 sector category of the relationship between m and x . Standard errors clustered by importer are in parentheses, with *** indicating 99% significance, ** indicating 95% significance, and * indicating 90% significance. Data years are 2001 through 2011.

Table 3
Relationship survival rates, global financial crisis.

Age	2004–2006	2007	Difference	Percent change
1	30.8	26.7	−4.1	−13.3
2	50.2	53.9	3.7	7.4
3	59.0	63.2	4.2	7.1
4	64.7	67.9	3.2	5.0
5	68.3	72.2	3.9	5.7
6	71.5	74.4	2.9	4.1
7	75.5	76.7	1.2	1.6
8	77.1	79.7	2.6	3.4

Note: This table documents facts about the survival of relationships between U.S. importers and foreign suppliers before and during the 2008 Global Financial Crisis. Column 2 shows the average survival rates for each age over the years 2004–2006 (where the survival rate in 2004 is the share of relationships alive in 2004 that are still alive in 2005, etc.), column 3 the survival rates in 2007, and columns 3 and 4 the absolute difference and percent change between these rates, respectively.

2.2.3. Within-relationship movements

Trade increases as relationships age. The differences between the count distributions and the value distributions in Panel A of Table 1 imply that, on average, older relationships trade more. We find one reason for this is that the amount traded within a surviving relationship increases over time. We set up the following regression specification to illustrate trade growth for relationships that were 10 years old in 2011:

$$Value_{mx,t} = \sum_{k=1}^K \beta_k \mathbb{1}[Age_{mx,t} = k] + f_{mx} + f_{mt} + u_{mx,t} \quad (1)$$

where $Value$ is the log value of trade between a U.S. importer m and a supplier x at time t , Age is the age of that relationship at time t (with $Age = 0$ - the first year a relationship is found - omitted), f_{mx} is a relationship fixed effect and f_{mt} is an importer-year fixed effect.

Table 4
Relationship survival rates, U.S.-China trade war.

Age	2014–2016	2017	Difference	Percent change
1	35.6	31.5	−4.1	−11.5
2	54.0	49.6	−4.4	−8.1
3	63.0	58.9	−4.1	−6.5
4	68.5	64.2	−4.3	−6.3
5	71.5	67.0	−4.5	−6.3
6	73.8	67.9	−5.9	−8.0
7	75.6	70.7	−4.9	−6.5
8	77.5	72.8	−4.7	−6.1

Note: This table documents facts about relationship survival of relationships between U.S. importers and Chinese suppliers before and during the 2018–2019 trade war between the United States and China. Column 2 shows the average survival rates for each age over the years 2014–2016 (where the survival rate in 2014 is the share of relationships in alive 2014 that are still alive in 2015, etc.), column 3 the survival rates in 2017, and columns 3 and 4 the absolute difference and percent change between these rates, respectively.

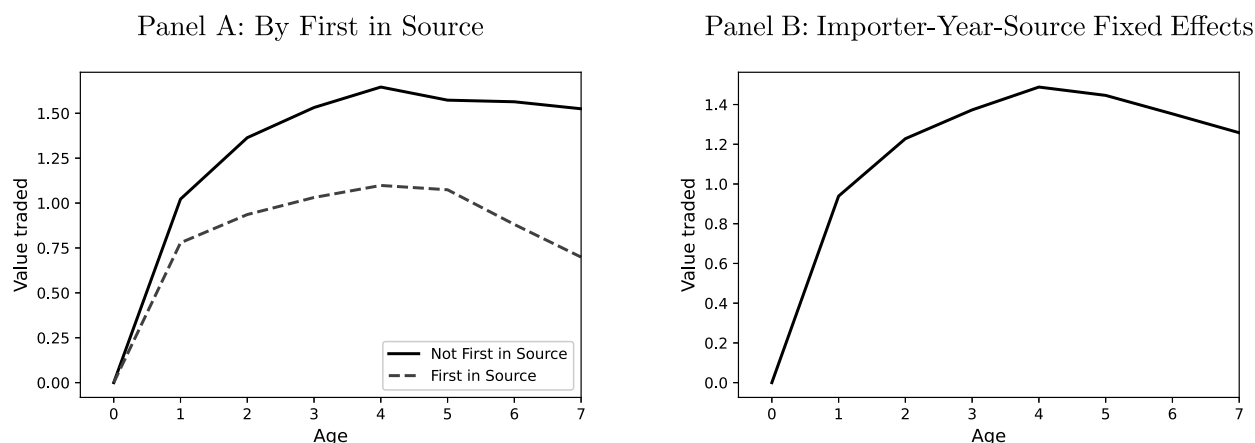


Fig. 4. Evolution of trade within a relationship. Note: This figure plots the coefficients for ages 1 through 7 from a regression of log imported value on relationship age dummies for relationships that were 10 years old in 2011 as in Eq. (1). The regression includes relationship fixed effects and importer-year fixed effects. Panel A separates out relationships that are not the first relationship for a U.S. importer with a particular source (solid line) from those that are the first relationship for a U.S. importer with a particular source (dashed line). Coefficients shown in Panel B are from a regression that includes importer-year-source fixed effects. All regressions uses data from 2001 through 2011.

The results of the regression are shown in Fig. 3 by graphing the β_k terms: as a relationship ages, the amount traded increases. As a relationship moves closer to its end, trade falls off, but for all relationships, trade initially grows over multiple years.¹⁵ This finding that trade grows within a relationship over time goes beyond earlier work, such as Bernard et al. (2018) and Carballo et al. (2018), that has shown a role for the extensive margin of importer–supplier pairs in cross-sectional data.¹⁶

This finding is robust to a number of other specifications. First, Fig. 4 Panel A augments the regression to include an interaction with whether or not the relationship is an importer’s first relationship in a particular source. Trade actually grows more in relationships that are not the first relationship in a source. The finding that the growth rate of imports for experienced importers is larger is somewhat surprising but clearly indicates that the relationship dynamics we uncover go over and above any dynamics at the importer–source level. Fig. 4 Panel B runs the same specification as our main specification, but includes importer–year–source fixed effects, which absorb any dynamics at the importer–source level, such as an importer increasing its purchases over time from a source. The pattern of trade growth remains quite strong, providing further evidence for the importance of relationship-level dynamics.

Prices decline in relationship age. We next look at the behavior of prices within relationships over time. To this purpose, we generate a dataset that is at the relationship–HS10 product level, as unit values are easier to interpret and study at that level of aggregation. By regressing the import price of a relationship–product on age dummies as in Eq. (1), we can plot the path of prices within relationships over time, controlling for relationship–product fixed effects. As shown in Fig. 5, prices decline with relationship age either when

¹⁵ Heise (2015) performs a similar exercise, also using the LFTTD data.

¹⁶ Importantly, this result is robust to accounting for “partial year effects”- work by Bernard et al. (2017) has shown firms that start exporting in January trade much more in a year than firms that start exporting in December, which can affect calculations of the intensive margin of trade.

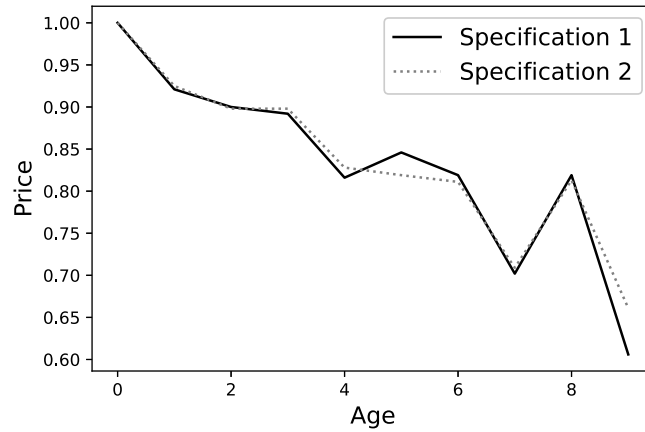


Fig. 5. Price regressions. Note: This figure plots the coefficients from two regressions of log unit values on relationship age dummies for relationships that were 10 years old in 2011. Specification 1 (the solid line) is based on a regression that includes relationship – HS fixed effects, importer-year, and supplier – year fixed effects. Specification 2 (the dashed line) is based on a regression that also includes source — HS fixed effects in addition to the fixed effects in Specification 1. Both regressions use data from 2001 through 2011.

controlling for importer-year and supplier-year fixed effects (solid line) or when further controlling for source-product-year fixed effects (dotted line). This pattern is consistent with a model of learning, as we illustrate later in the paper.

2.2.4. Heterogeneity in relationships from different sources

Cross-source differences in trade relationships are substantial. There are differences in the count distribution and value distribution across major import partners of the United States (see Table B.3). Some sources (such as China, Spain, and Brazil) have a low fraction of older relationships (age 3 or more) relative to all U.S. sources and have less than half of their exports to the United States in such relationships. Other sources (such as Germany, Japan, and Taiwan) have a much larger share of older relationships and have much more exports to the United States concentrated in such relationships.

Work such as [Martin et al. \(2020\)](#) has shown that relationship lengths are quite different across products. Interestingly, however, the differences in the age structure of trade we find do not reflect differences in product specialization. We show this in two ways. First, we directly control for industry effects to adjust the source-level count and value distributions, using the following specification:

$$Share_{isk} = \alpha_{ik} + \alpha_{is} + \varepsilon_{isk} \quad (2)$$

In this regression equation, $Share_{isk}$ is the share of relationships/trade in HS4 sector s from source i accounted for by relationships of age k , while the α_{ik} and α_{is} terms are source-age and source-sector fixed effects, respectively (see Tables B.4 and B.5). The relative ranking of exporters to the United States is little changed when using the industry-adjusted α_{ik} measure.¹⁷ Second, we check whether sources with more trade in old relationships in the aggregate also have more trade in old relationships at the sector level. More specifically, we calculate each source's overall share of trade in older relationships (as in Table B.3), as well as the shares for textiles (HS2 50-63), machinery and electrical goods (HS2 84-85), and footwear/headgear (HS2 64-67). We find that rank correlations are positive, implying that source orderings within industries are similar to the aggregate source ordering (see Table B.6).

Since heterogeneity in trade shares of relationships across sources is not driven by the sectoral composition of exports, in the theory section, we design a model that permits differential relationship dynamics across sources while abstracting from differences in exported products.

Source countries with better institutions have longer relationships. To shed light on which characteristics drive the observed heterogeneity in relationship lengths across sources, we run reduced-form regressions that control for importer and industry characteristics with the following specification:

$$RelLength_{mxc} = \beta_x Y_x + \beta_c Y_c + f_m + f_i + u_{mxc}, \quad (3)$$

where $RelLength$ is the age of a relationship in 2011 between U.S. importer m and supplier x in source c . Y_x is a set of supplier controls, including firm size (proxied by log total exports), and log total number of relationships. Regressions also include importer

¹⁷ Running a variant of Eq. (2) by dropping the k subscript and using either the share of long-term relationships or the value of trade in long-term relationships as the dependent variable also yields very similar α_i terms for cases where α_i fixed effects are or are not included.

Table 5
Determinants of importer–supplier relationship length.

Dependent variable: $RelLength_{msc}$							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$Size_x$	0.234*** (0.004)	0.234*** (0.004)	0.234*** (0.004)	0.234*** (0.004)	0.234*** (0.004)	0.234*** (0.004)	0.233*** (0.004)
$RelCount_x$	−0.191*** (0.013)	−0.191*** (0.013)	−0.191*** (0.013)	−0.191*** (0.013)	−0.191*** (0.013)	−0.191*** (0.013)	−0.191*** (0.013)
Source controls							
$RuleOfLaw_c$	0.059*** (0.009)	0.056*** (0.009)	0.058*** (0.009)	0.059*** (0.009)	0.056*** (0.009)	0.058*** (0.009)	0.078*** (0.010)
$PC - GDP_c$	0.042*** (0.009)	0.044*** (0.009)	0.042*** (0.009)	0.042*** (0.009)	0.044*** (0.009)	0.042*** (0.009)	−0.016 (0.012)
$Distance_c$	0.014*** (0.005)			0.013*** (0.007)			0.051*** (0.008)
$DistanceW_c$		0.020*** (0.007)			0.021*** (0.008)		
$DistanceCap_c$			0.016*** (0.006)			0.016*** (0.007)	
RTA_c				−0.004 (0.012)	0.003 (0.013)	−0.001 (0.012)	0.041*** (0.014)
$OECD_c$							0.181*** (0.018)
N	1,147,000	1,147,000	1,147,000	1,147,000	1,147,000	1,147,000	1,147,000

Note: This table presents results on the role of supplier factors and source institutions for the age of relationships. All regressions include importer fixed effects f_m and importer–industry fixed effects f_i — since importers can import from more than one industry, we assign each importer the HS2 industry that it imports the most from. The dependent variable is the age of a relationship in 2011, defined as the number of consecutive years a relationship has been found in the data, and is capped at 11. Controls at the supplier level include log total exports and log total number of relationships. Source controls are rule of law (from Kaufmann et al., 2010), per capita GDP, a trade agreement dummy (RTA_c), an OECD membership dummy, and distance from the U.S. measured in various ways. Data on the distance between country c and the United States comes from the CEPII gravity database and includes logged distance between most populated city ($Distance$), logged population-weighted distance between most populated cities ($DistanceW$), and logged distance between capitals ($DistanceCap$). Standard errors clustered by importer are in parentheses, with *** indicating 99% significance, ** indicating 95% significance, and * indicating 90% significance.

fixed effects f_m and importer–industry fixed effects f_i .¹⁸ Y_c is a set of source controls that includes rule of law, per capita GDP, distance from the U.S. measured in various ways, a trade agreement dummy, and an OECD membership dummy.¹⁹

Table 5 presents the results. The regressions show that U.S. importers have longer relationships with suppliers from sources with better institutions (higher rule of law). Per-capita GDP of the source is also positively correlated with relationship age, unless controlling for OECD membership. Interestingly, all distance measures are positively correlated with relationship length. This finding contrasts with results reported in Alborno et al. (2016), who find that firms survive less in markets farther away; the difference may be arising because of the different levels of aggregation between firm-level data and relationship-level data. Being in a trade agreement with the United States (RTA) is not significantly correlated with relationship age.

To sum up, in this section we show that most trade relationships are new while most trade occurs in long-term relationships. The survival probability of a relationship increases as a relationship ages and is affected by aggregate shocks. Trade within a relationship increases as the relationship ages, while prices decline. Finally, there is notable heterogeneity in the distributions of relationships across sources. Guided by these facts, we develop a model of relationship dynamics, calibrate its parameters source by source, and use the model to calculate the value of relationships to importers and to the economy.

3. Model

This section presents a two-country trade model that adds importer–supplier relationship dynamics to the standard setup in Krugman (1980).²⁰ Relationship dynamics are modeled in reduced form, delivering two sets of country-specific moments: average amounts traded by relationship age and conditional survival rates of relationships from period t to $t + 1$. As we showed above, in the data, both sets of moments tend to increase with relationship age.

The relationship dynamics are consistent with several mechanisms. Average revenues can increase in relationship age as a result of learning-by-doing or relationship-specific investments leading to higher success probabilities and more sales within a relationship

¹⁸ Since importers can import from more than one industry, we assign each importer the HS2 industry that it imports the most from.

¹⁹ Rule of law data comes from Kaufmann et al. (2010), while other variables come from the CEPII gravity database (Conte et al., 2022). The distance between source c and the United States includes logged distance between most populated city ($Distance$), logged population-weighted distance between most populated cities ($DistanceW$), and logged distance between capitals ($DistanceCap$).

²⁰ While the Krugman model does not feature firm heterogeneity, the GE analysis allows to study the entry dynamics and the aggregate response of the economy in response to shocks in a parsimonious way. Future work could extend our setup to a setting with heterogeneous firms and endogenous exit.

over time. Or, there may be a selection effect due to learning about the reliability of suppliers over the course of a relationship. That is, unreliable suppliers get dropped over time, improving the average quality of the remaining suppliers.²¹ Conditional survival rates may change with age for the same reasons. We do not take a stance on which of these mechanisms drives the empirical patterns described above. However, in Section 3.4, we briefly outline one potential micro-foundation, a learning model based on Araujo et al. (2016).

3.1. Setup

There are two countries i and j . An upstream firm (the supplier) in country i can directly sell to domestic consumers. However, when exporting, the upstream firm has to sell through a downstream firm (the distributor) in country j . The supplier has all bargaining power and receives the full surplus from exporting. Production requires c units of labor per unit of output with wage w . When exporting, the supplier faces an iceberg-type trade cost $\tau > 1$, that is, she needs to ship τ units in order for 1 unit to arrive in the final destination. Assume that for domestic sales, there are no iceberg costs (that is, $\tau = 1$).

For exporting, we assume that the supplier and the distributor build relationship-specific capital over time, making longer-lasting relationships more likely to survive. We define the success probability of a relationship of age k to be Θ_k . This gives rise to the following operating profits from exporting for a supplier in an age- k relationship:

$$E[\pi_k] = \Theta_k p(\Theta_k) q(\Theta_k) - \tau q(\Theta_k) c w, \quad (4)$$

where $p(\Theta_k)$ is the price paid by the distributor to the supplier, $q(\Theta_k)$ is the quantity sold, and w is the wage. The distributor pays the production costs to the supplier before knowing whether the transaction is successful. Relationships die after reaching the maximum relationship age K .²²

We assume that domestic sales are always at their first-best level (as if, for domestic sales, $\Theta_k = 1$).²³ In addition, assume that suppliers need to sell domestically and export in order to survive. That is, when an export relationship ends, this forces the supplier to exit.²⁴ Further, assume that acquiring a new supplier requires f_E units of labor. Let p^{ii} refer to the domestic price in country i and p^{ij} refer to the price paid in country j for a good from country i .

Demand and supplier profits. Sales to final consumers take place under monopolistic competition and final consumers have standard CES preferences given by: $Q_t = \left(\int_{\omega \in \Omega} q_t(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}}$ over varieties $\omega \in \Omega$, with the ideal price index $P_t = \left(\int_{\omega \in \Omega} p_t(\omega)^{1-\sigma} d\omega \right)^{\frac{1}{1-\sigma}}$. Aggregate demand is then given by $A_t = P_t^\sigma Q_t$, while demand for variety ω is $q_t(\omega) = A_t p_t(\omega)^{-\sigma}$. Then, the expected profits of a supplier at time t in country i that sells domestically and has a k -year old international trade relationship with a distributor in country j are:

$$E[\pi_{t,k}^i] = p_t^{ii} q_t^{ii} + \Theta_k^i p_{t,k}^{ij} q_{t,k}^{ij} - q_t^{ii} c^i w_t^i - \tau q_{t,k}^{ij} c^i w_t^i, \quad (5)$$

Note that export prices and quantities depend on relationship age k while domestic prices and quantities are independent of k , as only international sales are subject to the relationship dynamics described above. The first-order conditions of profit maximization imply:

$$p_t^{ii} = \frac{\sigma}{\sigma-1} c^i w_t^i \quad (6)$$

$$p_{t,k}^{ij} = \frac{\tau}{\Theta_k^i} p_t^{ii} \quad (7)$$

From this, we can derive overall expected profits as:

$$E[\pi_{t,k}^i] = \frac{1}{\sigma} (p_t^{ii})^{1-\sigma} \left(A_t^i + \tau^{1-\sigma} (\Theta_k^i)^\sigma A_t^j \right). \quad (8)$$

With expected (operating) profits from exporting:

$$\begin{aligned} E[\pi_{t,k}^{ij}] &= \Theta_k^i p_{t,k}^{ij} q_{t,k}^{ij} - \tau q_{t,k}^{ij} c^i w_t^i \\ &= \frac{\Theta_k^i}{\sigma} (p_{t,k}^{ij})^{1-\sigma} A_t^j = \frac{\Theta_k^i}{\sigma} \left(\frac{\tau}{\Theta_k^i} p_t^{ii} \right)^{1-\sigma} A_t^j \end{aligned} \quad (9)$$

Eq. (9) makes clear that the delivery probability, Θ_k^i , affects expected profits from exporting via two channels. First, there is a direct effect, as a sale is more likely to succeed with a higher Θ_k^i . Second, there is an indirect effect: the optimal price $p_{t,k}^{ij}$ decreases in

²¹ A third explanation is a different type of selection effect in which relationships with initially higher revenues tend to survive. While this type of selection effect likely plays a role for the aggregate patterns that we uncover, on its own it cannot rationalize our findings, as our results on the increase of trade with relationship age include relationship fixed effects.

²² We need to set a maximum age K in order to later solve the model numerically.

²³ Of course, even for domestic sales revenues may be dynamic, for example through building customer capital as in Drozd and Nosal (2012), or because the supplier may still rely on some local distributor. The key assumption here is that established relationships are more important for international trade than for domestic sales, for example, because communication and travel costs are much higher. For tractability, we focus on the case of no domestic friction, but it would be straightforward to relax that assumption.

²⁴ This assumption is made for tractability. It could, in principle, be relaxed but would require rewriting and solving the model recursively.

Θ_k^i , leading to a larger order and hence, larger revenues and profits if the trade is successful. A successful international transaction generates revenues of:

$$R_{i,k}^{ij} = p_{i,k}^{ij} q_{i,k}^{ij} = A_i^j \left(\frac{\sigma}{\sigma-1} \frac{\tau c^i w_t^i}{\Theta_k^i} \right)^{1-\sigma} \quad (10)$$

In the calibration section, we use the average amounts traded by relationship age as target moments. More specifically, we target revenues at age k normalized by revenues of a new relationship to get:

$$\hat{R}_k^{ij} = \frac{R_k^{ij}}{R_0^{ij}} = \left(\frac{\Theta_k^i}{\Theta_0^i} \right)^{\sigma-1}. \quad (11)$$

Note that the ratio $\hat{R}_k^{ij}$ has no t subscript, as we calibrate the model to the steady state. Also note that the ratio does not depend on any general equilibrium objects like the aggregate price level or aggregate demand as these cancel out. Instead, it is completely determined by the different values of Θ_k^i for different ages. Finally, as discussed in Appendix D, we take “partial year effects” into account by reducing trade in the first year to reflect the fact that relationships may only trade in parts of the first year of their existence (Bernard et al., 2017).

3.2. The value of a relationship

Define the unconditional probability that a relationship between a supplier in i and a distributor in j survives to age k as $alive_k^{ij}$ ²⁵:

$$alive_k^{ij} = \begin{cases} 1 & k = 0 \\ \prod_{r=1}^k \Theta_{r-1}^i & k \geq 1 \end{cases} \quad (12)$$

This condition can also be written as an auto-regressive process:

$$alive_k^{ij} = \Theta_k^i alive_{k-1}^{ij}, \quad (13)$$

with $alive_0^{ij} = 1$.

Now, define the value of a relationship of age k as the sum of expected profits from being in that relationship relative to the sum of expected profits from being in a new relationship.²⁶ Note that we do not explicitly assume that suppliers discount future profits. This is not necessary, as the survival probability, Θ_k^i , is always smaller than one.²⁷ Using Eq. (9), the expected profits of a newly formed relationship between a supplier in i and a distributor in j can be derived as:

$$\begin{aligned} \Pi_0 &= \sum_{s=0}^K alive_s^{ij} E[\pi_s^{ij}] \\ &= \frac{1}{\sigma} (\tau p^i)^{1-\sigma} A^j \sum_{s=0}^K alive_s^{ij} (\Theta_s^i)^\sigma, \end{aligned} \quad (14)$$

To compute this object for existing relationships, we note that the probability a relationship of age k survives to age $k+s$ is $alive_{k+s}^{ij}/alive_k^{ij}$. Again calculating the sum of expected profits of a relationship, this time for age k , gives:

$$\begin{aligned} \Pi_k &= \sum_{s=0}^{K-k} \left(\frac{alive_{k+s}^{ij}}{alive_k^{ij}} \right) E[\pi_{k+s}^{ij}] \\ &= \frac{1}{\sigma} (\tau p^i)^{1-\sigma} A^j \sum_{s=0}^{K-k} \left(\frac{alive_{k+s}^{ij}}{alive_k^{ij}} \right) (\Theta_{k+s}^i)^\sigma. \end{aligned} \quad (15)$$

Then, the value of a relationship of age k is:

$$V_k = \frac{\Pi_k}{\Pi_0} = \frac{\sum_{s=0}^{K-k} \left(\frac{alive_{k+s}^{ij}}{alive_k^{ij}} \right) (\Theta_{k+s}^i)^\sigma}{\sum_{s=0}^K alive_s^{ij} (\Theta_s^i)^\sigma}. \quad (16)$$

²⁵ In this section, we drop the t subscript, as we focus on the steady-state distributions. We reintroduce the time dimension below when solving for the dynamic transition path in general equilibrium.

²⁶ We do not assess the value of not being in a relationship, meaning our definition differs somewhat from that in the search-based framework of Krolkowski and McCallum (2021). Instead, we assume that distributors or suppliers who are unable to match simply drop out and have zero profits.

²⁷ Of course, it would be straightforward to introduce an additional discount factor. This would, however, not affect any of our key results.

3.3. Solving for the equilibrium

There are five equilibrium conditions. For each country, there is a free entry condition and a labor market clearing condition. In addition, the balanced trade condition has to hold. Finally, the equilibrium has to be consistent with the law of motion for the mass of relationships of age k that transition to age $k + 1$. To keep the equations tractable, it is useful to introduce some notation. We shorten the equation for domestic profits to:

$$\pi_t^{ii} = \frac{1}{\sigma} (p_t^{ii})^{1-\sigma} A_t^i = \frac{r_t^{ii}}{\sigma} A_t^i, \quad (17)$$

with $r_t^{ii} = (p_t^{ii})^{1-\sigma} = \left(\frac{\sigma}{\sigma-1} c^i u_t^i\right)^{1-\sigma}$. Since domestic deliveries are always successful and there are no dynamics for domestic sales, there is no expectation operator and no relationship age subscript k . We shorten expected profits from exporting in Eq. (9) to:

$$E[\pi_{t,k}^{ij}] = \frac{\theta_k^{ij}}{\sigma} (p_{t,k}^{ij})^{1-\sigma} A_t^j = \frac{r_{t,k}^{ij}}{\sigma} A_t^j, \quad (18)$$

with $r_{t,k}^{ij} = \theta_k^{ij} (p_{t,k}^{ij})^{1-\sigma} = \theta_k^{ij} \left(\frac{\tau}{\theta_k^{ij}} p_t^{ii}\right)^{1-\sigma}$. Unlike the domestic profit condition, this equation depends on the age of the relationship k . Finally, call the mass of relationships in country i at time t of age k $N_{t,k}^i$. This means that the number of new relationships in country i at time t is $N_{t,0}^i$.

Free entry. The free entry condition must hold in both countries, such that the fixed cost of entering production is equal to the expected sum of future profits of an entrant at time t .²⁸ We have:

$$w_t^i f_E = \sum_{s=0}^K alive_s^{ij} (\pi_{t+s,s}^{ii} + E[\pi_{t+s,s}^{ij}]), \quad (19)$$

where $alive_k^{ij}$ is the cumulative probability that a trade relationship survives k years, as previously. Now, replacing profits by Eqs. (17) and (18), we can rewrite to:

$$w_t^i \sigma f_E = \sum_{s=0}^K alive_s^{ij} (r_{t+s,s}^{ii} A_{t+s}^i + r_{t+s,s}^{ij} A_{t+s}^j) \quad (20)$$

Isolating the first term of the sum, combining the two versions of (20) for w^i and w^j , and then solving for demand in i delivers:

$$A_t^i = \frac{r_t^{ii} (w_t^i \sigma f_E - \Pi_{t,1+}^i) - r_{t,0}^{ij} (w_t^j \sigma f_E - \Pi_{t,1+}^j)}{r_t^{ii} r_{t,0}^{jj} - r_{t,0}^{ij} r_t^{ji}}, \quad (21)$$

with $\Pi_{t,1+}^i = \sum_{s=1}^K alive_s^{ij} (r_{t+s,s}^{ii} A_{t+s}^i + r_{t+s,s}^{ij} A_{t+s}^j)$ the sum of expected profits from period 1 onward, and an analogous equation for demand in j .²⁹

Labor market clearing. Second, labor markets in both countries need to clear, such that the labor used for entry and to generate the output of all relationships is equal to the total labor endowment. Denote total labor in country i as L^i . Then, labor market clearing requires:

$$L^i = N_{t,0}^i f_E + \sum_{s=0}^K N_{t,s}^i c^i \left((p_t^{ii})^{-\sigma} A_t^i + \tau (p_{t,s}^{ij})^{-\sigma} A_t^j \right) \quad (22)$$

Solving for entry in period t by isolating the first two terms on the right-hand side delivers:

$$N_{t,0}^i = \frac{L^i - L_{P+,t}^i}{f_E + c^i \left((p_t^{ii})^{-\sigma} A_t^i + \tau (p_{t,0}^{ij})^{-\sigma} A_t^j \right)}, \quad (23)$$

with $L_{P+,t}^i = \sum_{s=1}^K N_{t,s}^i c^i \left((p_t^{ii})^{-\sigma} A_t^i + \tau (p_{t,s}^{ij})^{-\sigma} A_t^j \right)$ the production labor used by all suppliers that are not new in country i .

Balanced trade. We close the model by imposing balanced trade, which has to hold because we assume that countries are in financial autarky.³⁰ That is the total value of country i 's imports needs to equal the total value of country i 's exports:

$$\sum_{s=0}^K (N_{t,s}^j r_{t,s}^{j,i} A_t^i - N_{t,s}^i r_{t,s}^{i,j} A_t^j) = 0 \quad (24)$$

²⁸ As in Krugman (1980), we do not assume any fixed cost of exporting, so that any supplier that enters also exports.

²⁹ The entire set of equilibrium conditions for both i and j are written out in Appendix C.

³⁰ We assume financial autarky for tractability. While this is beyond the scope of our paper, it would be fascinating to study how dynamics would change in our setup if countries had the ability to borrow from another, as e.g. in Reyes-Heroles (2016).

Law of motion for number of relationships. Finally, the law of motion of the number of relationships is directly pinned down by the success probability:

$$N_{t+1,k+1}^i = \Theta_k^i N_{t,k}^i \quad (25)$$

Solving the dynamic model. Normalizing the wage in country i to 1 and given a success probability Θ_k^i for all ages k and all countries i (which will be calibrated from the moments summarized in Section 2), we use these equilibrium conditions to solve for the unknowns $\{A_t^i, A_t^j, N_{t,k}^i, N_{t,k}^j, w_t^j\}$ in every period t and for all ages k . We solve for the dynamic transition path of the model (starting from any distributions of relationships at home and abroad $N_{0,s}^i, N_{0,s}^j$, for $s = 1, \dots, K$) in the following way:

1. For periods $t = 1$ to n , guess a sequence of demand at home and abroad, A_t^i, A_t^j , and a sequence of the foreign wage w_t^j . For all numerical analysis presented later, we use the steady state values as initial guesses, which works well.
2. Repeat until convergence (of w_t^j, A_t^i and A_t^j):
 - (a) Take the sequences $w_0^j, \dots, w_n^j; A_1^i, \dots, A_n^i$ and $A_1^j, \dots, A_n^j$ as given and search for the relative wage w_0^j that minimizes the balanced trade condition (Eq. (24)). To do so:
 - i. For given w_0^j , solve for demand levels in period 0, A_0^i and A_0^j , using Eq. (21).
 - ii. Then, solve for entry levels, $N_{0,0}^i$ and $N_{0,0}^j$, using Eq. (23).
 - iii. Evaluate Eq. (24) and adjust the estimate of w_0^j .
 - (b) Based on entry in period 0, calculate new relationship counts for period 1, using the law of motion Eq. (25). Then solve for demand levels, A_1^i and A_1^j , entry levels $N_{1,0}^i$ and $N_{1,0}^j$ and relative wage w_1^j for period 1, following the same steps as under (a).
 - (c) Repeat for all periods 2 to n
 - (d) Update the guesses for the sequences w_t^j, A_t^i and A_t^j
 (we use $A_t^{i,U\text{pdated}} = 0.95A_t^i + 0.05A_t^{i,Guess}$ for all $t = 1$ to n)

We set $\sigma = 3$ and $c = 1$ for our baseline analysis but provide robustness with alternative values of σ . Note that to calculate entry at period t , we need to know the demand and relative wage in all periods up to $t + K$. We set the demand and relative wage for all periods after period n to their steady-state values. In our numerical exercises, we set the maximum number of periods to $n = 100$ to make sure that the economy converges back to steady state well before period n , as well as the maximum age of a relationship to $K = 200$. We further set the partial year factor to 0.5, τ to 1.1, L and L^* to 1000, and f_E to 0.01. More details about the full set of equilibrium conditions and the steady state solution of the model can be found in Appendix C.

Aggregate price indices and aggregate quantities. After solving for the equilibrium as above, we can generate the other aggregate objects in the model. The price index at time t is an aggregation of prices paid for all goods successfully supplied at time t , taking into account that delivery is only successful with probability Θ_k . Thus, we can derive the ideal price index as:

$$P_t^i = \left(\int_{\omega \in \Omega} p(\omega)^{1-\sigma} d\omega \right)^{\frac{1}{1-\sigma}} = \left((p_t^{ii})^{1-\sigma} \sum_{s=0}^K N_{t,s}^i + \sum_{s=0}^K \Theta_s^j N_{t,s}^j (p_t^{ji})^{1-\sigma} \right)^{\frac{1}{1-\sigma}} \quad (26)$$

We can also derive the aggregate consumption bundle as:

$$Q_t^i = \left(\int_{\omega \in \Omega} q(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}} = \left((q_t^{ii})^{\frac{\sigma-1}{\sigma}} \sum_{s=0}^K N_{t,s}^i + \sum_{s=0}^K \Theta_s^j N_{t,s}^j (q_t^{ji})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} \quad (27)$$

This completes the description of the general equilibrium model for a given set of parameters $\{\Theta_k^i\}, k \in [0, \dots, K]$.

3.4. Micro foundation: A learning model

Before moving to the main calibration and counterfactual analysis, we outline one possible micro-foundation based on the learning mechanism in Araujo et al. (2016). We derive the model, calibrate it, and illustrate how it aligns with our empirical findings on prices, quantities and relationship survival.

The learning model is consistent with several empirical patterns documented in this paper, in particular, the decline of unit values within relationships over time and the correlation between relationship age and the rule of law at the country level.³¹ In addition, research on payment terms in international trade (Antràs and Foley, 2015; Garcia-Marin et al., 2019) and research on relationships in developing economies (Macchiavello and Morjaria, 2015) also suggests a central role for learning about the reliability of trading partners.³²

³¹ Araujo et al. (2016) and Aeberhardt et al. (2014) provide evidence consistent with a role for learning and enforcement frictions in international trade, using Belgian and French firm-level trade data, respectively.

³² In particular, Antràs and Foley (2015) and Garcia-Marin et al. (2019) show that firms switch payment terms within relationships over time, in particular from cash-in-advance to open account transactions. Open account exposes suppliers to substantial risks and the observed switches are difficult to rationalize without learning by suppliers about their distributors.

A distributor is matched randomly with a supplier. There are two types of suppliers. With probability $\hat{\theta}$, a supplier is reliable. A reliable supplier always successfully delivers the goods to the distributor. With probability $1 - \hat{\theta}$, the supplier is unreliable, meaning that it tries to deviate from the contract when there is an opportunity to do so and not deliver the goods.³³ If a supplier tries to deviate from the contract, it is forced to fulfill it with exogenous probability λ , a parameter that is related to the overall quality of institutions in the supplier country.³⁴ Finally, in order to ensure that perfect relationships do not dominate outcomes, an exogenous probability of relationship death δ also needs to be introduced.

The timing is as follows. First, the distributor pays the supplier. Importantly, the payment takes place before the distributor receives the goods.³⁵ In addition, there is imperfect contract enforcement. That is, the distributor cannot punish the supplier for not delivering the goods (or for delivering goods of inferior quality) over and above ending the relationship. Second, the supplier delivers the goods and the distributor sells them to consumers. Note that the distributor receives the goods either if the supplier is reliable or if supplier is unreliable but forced to deliver the goods. Finally, the distributor decides whether to continue transacting with the supplier. A relationship will only be continued if the prior transaction was successful.

Bayesian updating. As there are two types of suppliers in the economy, learning plays a central role. Initially, distributors believe (correctly) that the probability the supplier is reliable is equal to the population mean, $\hat{\theta}$. Every period that a relationship survives, distributors update their beliefs about their supplier according to Bayes' rule. If a distributor has successfully purchased from a supplier for k periods, the posterior probability that the supplier is reliable can be derived as:

$$\theta_k = \frac{\hat{\theta}}{\hat{\theta} + (1 - \hat{\theta})\lambda^k}. \quad (28)$$

Importantly, the probability only changes with the length of time that a distributor has been buying from the same supplier. It is easy to see that for large k , θ_k converges to 1; that is, the distributor is almost certain that the supplier is reliable.

The success probability. Consider again a relationship of age k . The distributor will receive the goods from the supplier next period under two scenarios: either the supplier is reliable (an event with expected probability θ_k), or the supplier is unreliable but is forced to deliver the goods anyways (an event with expected probability $(1 - \theta_k)\lambda$). Thus, after k successful transactions, the success probability is:

$$\begin{aligned} \Theta_k &= (\theta_k + (1 - \theta_k)\lambda) = \left(\frac{\hat{\theta}}{\hat{\theta} + (1 - \hat{\theta})\lambda^k} \right) (1 - \lambda) + \lambda, \\ &= \frac{(1 - \hat{\theta})\lambda^{k+1} + \hat{\theta}}{(1 - \hat{\theta})\lambda^k + \hat{\theta}}. \end{aligned} \quad (29)$$

This expression for the success probability Θ_k is increasing in k , the age of the relationship, as well as in $\hat{\theta}$, the fraction of reliable suppliers in the source country.³⁶ There are two competing effects of the parameter λ on the delivery probability. Countries that have better enforcement (a higher λ) have a higher initial success probability Θ_0 , but slower learning, meaning smaller increases in the success probability as k increases.

Relationship survival, average age, and institutional quality. A relationship of age k survives for another period if it is not hit by the exogenous death shock and if there is a successful delivery. It therefore survives with (conditional) probability $(1 - \delta)\Theta_k$.

The unconditional probability that a relationship is still alive after k periods is the probability it has survived all previous periods: $alive_k = alive_{k-1}(1 - \delta)\Theta_{k-1}$. Since the probability of any new relationship being alive at least 0 periods is 1 (that is, $alive_0 = 1$), this delivers:

$$alive_k = (1 - \delta)^k (\lambda^k(1 - \hat{\theta}) + \hat{\theta}). \quad (30)$$

From this, we can derive the average relationship age as:

$$age = \frac{\sum_{k=0}^{\infty} alive_k k}{\sum_{k=0}^{\infty} alive_k} = \frac{\sum_{k=0}^{\infty} (1 - \delta)^k (\lambda^k(1 - \hat{\theta}) + \hat{\theta}) k}{\sum_{k=0}^{\infty} (1 - \delta)^k (\lambda^k(1 - \hat{\theta}) + \hat{\theta})} \quad (31)$$

³³ Araujo et al. (2016) show how these firm types can be further micro-founded by assuming that reliable firms discount the future by less than unreliable firms. That is firms are reliable if they are patient and unreliable if they are not.

³⁴ Of course, λ can be interpreted more broadly. For example, it could reflect the probability that a supplier gets an opportunity to cheat rather than actual contract enforcement. Moreover, deviating from the contract does not necessarily mean that the supplier does not send any goods. For example, an unreliable supplier may send goods of inferior quality or with a long delay. In that case, λ could also be interpreted more broadly as the overall quality of a supplier's workforce, which may facilitate the provision of quality goods, while $\hat{\theta}$ could be interpreted as the effectiveness of a supplier's manager.

³⁵ That is, the distributor has to fully pay before delivery — cash-in-advance payment. Or, the distributor may pay after delivery but without having sufficient time to verify the quality of the goods before paying. This assumption can be relaxed and analysis is available upon request.

³⁶ $\frac{\partial}{\partial \hat{\theta}} \left(\frac{\hat{\theta}}{\hat{\theta} + (1 - \hat{\theta})\lambda^k} \right) = \frac{\lambda^k}{(\hat{\theta} + (1 - \hat{\theta})\lambda^k)^2} > 0$.

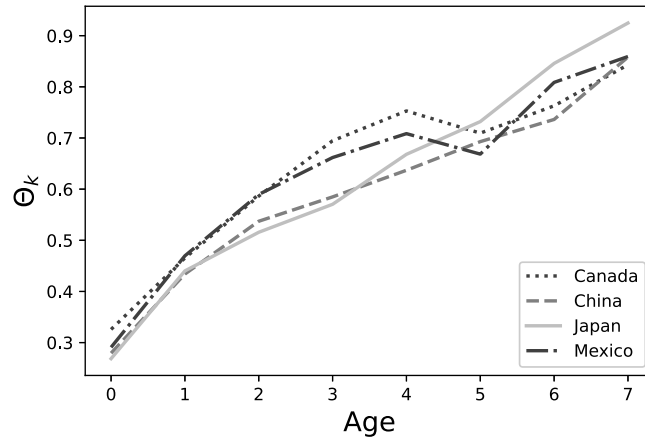


Fig. 6. Calibrated Θ_k for 4 major trading partners. Note: This figure shows the calibrated values for Θ_k^i for relationships between U.S. importers and foreign suppliers from 4 major import sources of the United States.

Echoing our empirical result from Table 5 that countries with better institutional quality tend to have older relationships, we can derive the following proposition:

Proposition 1. *A country's average relationship age increases in the share of reliable suppliers in a country $\hat{\theta}$, and in the enforcement probability λ , for $\lambda > 1 - \frac{\delta}{(1-\delta)\sqrt{\hat{\theta}}}$.*

Proof. See Appendix E .

Trade relationships, on average, last longer if there is a larger fraction of reliable suppliers to begin with. In addition, for sufficiently large λ , the average age also rises with the strength of enforcement. The relationship between these parameters and the model-based average age of a relationship is plotted in Figure A.1 .

4. Calibration and counterfactuals

In this section, we calibrate and evaluate the model. Section 4.1 presents the calibration of Θ_k . Section 4.2 calculates the value of relationships to importers by source based on the calibrated parameters. Section 4.3 uses the full general equilibrium model to quantify how shocks to the distribution of relationships affect efficiency, trade flows, and consumption.

4.1. Matching with the data

Our baseline calibration of Θ_k^i targets two sets of dynamic moments: the conditional survival rates of relationships from periods t to $t+1$ (Θ_k^i in the model) and trade revenues relative to the initial year by relationship age (a function of the ratio of different Θ_k^i , derived in Eq. (11)). We employ an algorithm that searches for the parameter vector $\{\Theta_k^i\}_{k=0}^7$ by minimizing the sum of squared differences between the moments in the data and those predicted by the model. Specifically, we solve the following problem:

$$\arg \min_{\Theta_0^i, \dots, \Theta_7^i} (1-\beta) \sum_{k=0}^7 (\Theta_k^i - \hat{\Theta}_k^i)^2 + \beta \sum_{k=0}^7 (\hat{R}_k^i - \hat{\hat{R}}_k^i)^2 \quad (32)$$

where $\hat{R}_k^i$ and Θ_k^i are the values predicted by the model, and $\hat{\hat{R}}_k^i$ and $\hat{\Theta}_k^i$ are taken from the data. We set $\beta = 0.5$.

We run this procedure for 10 major trading partners of the United States, indexed by i , as well as for the aggregate of all U.S. import sources ("World").³⁷ For each country, we use information on eight cohorts — that is, on relationships that are zero to seven years old. To generate the trade revenue moments, we first calculate the average growth rate of trade for relationships of age k from 2010 to 2011. This information allows us to compute $\hat{\hat{R}}_k^i$, the level of trade relative to the initial year for all relationship ages. To obtain the survival rates, we calculate the share of relationships of age k in 2010 that survived to 2011. The 8 parameters for each country are therefore selected by matching 16 moments in the model with their empirical counterparts.

Calibrated values of the success probability Θ_k^i for major trading partners of the United States are illustrated in Fig. 6 (exact values are reported in Table B.7). Consistent with the underlying survival probabilities and traded values, the probability of a successful

³⁷ The 10 major trading partners are Canada, China, Germany, France, India, Japan, Korea, Mexico, Taiwan, and the United Kingdom.

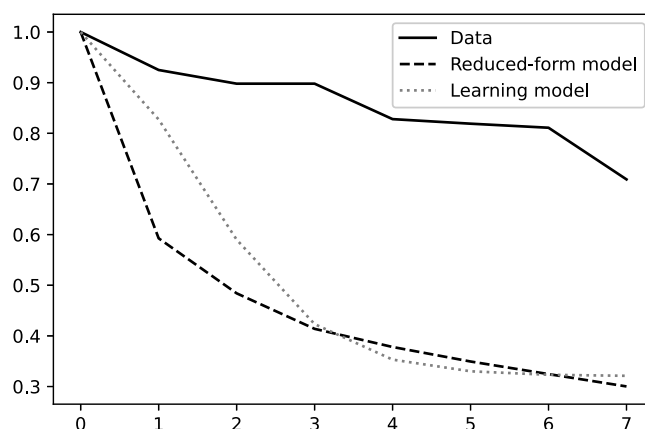


Fig. 7. Price predictions: Model vs. data. This figure compares predicted price paths for different model calibrations with the estimated path of unit values within relationships. The black dashed line shows the price path implied by the baseline reduced-form model calibration. The dotted gray line shows the price path implied by the learning model calibration.

relationship increases in the age of a relationship, with a substantial amount of heterogeneity across trading partners. As we show below, this variation in Θ_k^i generates differences across source in the value of relationships as well as the response of trade flows to disruptions in relationships.

Model fit. The calibrated model outcomes line up well with the underlying data (as shown in Figure A.2). While one may expect the flexible model with 8 parameters to do a good job in matching the 16 moments per country, it is notable that, for a given country-age cohort, a single parameter is able to simultaneously match the conditional survival probability as well as the average value of trade. The good fit supports the assumption that traded quantities are related to the expected success probability as assumed in the model.

We can also check how the model matches the cross-sectional count and value in 2011, which are not targeted in the calibration. In general, the distributions generated from the model and in the data are quite similar, supporting the assumption that trade relationship distributions are close to their steady state (see Figure A.3).³⁸

Price predictions. By calibrating the learning model in a similar fashion as above, we can revisit the empirical findings on prices within a relationship over time, shown in Fig. 5. For this, we additionally calibrate the three parameters of the learning model, δ , λ , and $\hat{\theta}$ to the sixteen dynamic moments for the “World” aggregate, which is done analogously to the calibration of the reduced form model. Note that the learning model is hence notably more parsimonious than the reduced-form model (three vs. eight parameters).

We then calculate within-relationship price paths using Eq. (7) for the learning model with the implied Θ_k from Eq. (29) as well as for our baseline reduced-form model, and plot them against the empirical observed path of unit values. This is shown in Fig. 7. It can be seen that the price path of the parsimonious learning model is relatively similar to that of the reduced-form model, indicating that the learning model does a reasonable job at capturing the general dynamics of the baseline reduced-form model.³⁹ Both calibrations also generate declines in prices over the course of a relationship that are qualitatively consistent with the estimated path of unit values within relationships over time in the data.

4.2. The value of trade relationships to firms

Next, we quantify the value of relationships using the estimated parameters and Eq. (16). As we only have disclosed moments for relationships up to age 7 for each country, we need to take a stance on the evolution of Θ_k for older relationships. In our baseline, we set Θ_k to the level of a relationship of age 7 for all relationships of age $k = 8$ and higher. We compute the value of age 8 relationships across sources to illustrate the value of “old” relationships. Table 6 presents our results.

According to our estimates, old relationships are most valuable relative to new relationships in Korea (where an established relationship has a sum of expected profits 72 times higher than a new relationship) and Japan (44 times higher). Across all countries, the average expected profits from a relationship of age 8 are 21 times higher than those from a new relationship. The large size of these estimates should not be surprising, given our empirical findings that long-term relationships foster higher trade as well as greater survival.

³⁸ The model explains 95 percent of the cross-sectional variation. Panel A of Figure A.4 shows a scatterplot of all 160 observations of the cross-sectional distributions (10 countries with 8 age-cohorts of count and value shares) in the data against the model-generated moments. Panel B shows that the model matches the targeted dynamic moments (survival rates and growth rates of trade) for the entire set of countries well, too.

³⁹ The learning model also has a good overall fit to the targeted moments.

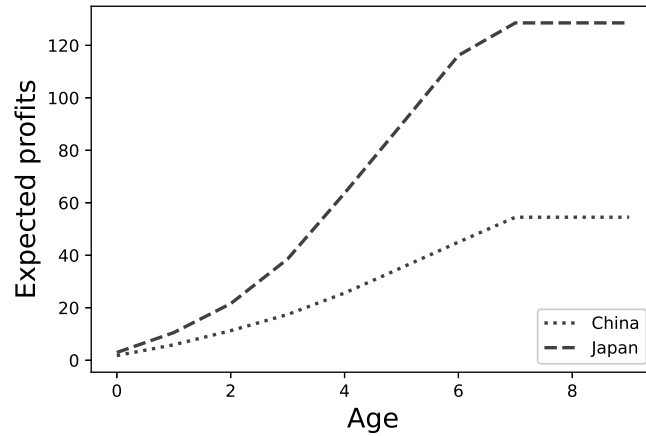


Fig. 8. Expected profits and relationship age. Note: This figure plots the sum of expected profits for a relationship of age k for China and Japan, based on each country's calibrated Θ_k^i and Eq. (15).

Table 6
The value of relationships.

Country	New	Old	Ratio
Canada	3.2	46.7	14.4
China	1.8	54.5	30.6
Germany	3.1	78.4	25.3
France	4.4	21.2	4.8
India	2.7	67.6	25.3
Japan	2.9	128.6	43.6
Korea	1.8	131.6	71.9
Mexico	2.8	55.2	19.7
Taiwan	2.8	59.4	21.3
United Kingdom	3.2	21.7	6.8

Note: Columns (1) and (2) of this table present the sum of expected profits from a new and an old relationship, respectively. Column (3) reports the ratio of the sum of expected profits from an old relationship over the sum of expected profits from a new relationship — what we call the value of a relationship. A new relationship is the first time two firms interact and an old relationship is a relationship of age $k \geq 8$, after Θ_k^i has reached its long-term value.

By accounting for the full set of survival and trade-by-age profiles across source countries, the model delivers insights about the value of relationships to firms that are not available from a straight read of the data. For example, countries with higher relationship values are not simply those with fewer long-term relationships: both Korea and China have nearly identical shares of long-term relationships (as shown in Table B.3), but they differ greatly in the relative value of an old relationship (72 versus 31), as old relationships in Korea both trade more and have higher survival rates. The reverse holds as well: sources with similar values of old relationships need not have similar age distributions. Table 6 shows that old relationships with Mexican and Taiwanese suppliers are similarly valuable, even though in the data, the shares of old relationships and the amount traded within old relationships are very different in these countries. The model-based value calculation accounts for both the rising conditional survival probability and the growth of trade as a relationship ages, both of which vary across countries.

There are two key determinants of the value of a relationship. First, the level of Θ_k across ages determines the average remaining life expectancy of a relationship of age k . Second, the “slope” of Θ_k over different relationship ages determines the increase in survival and trade volumes as a relationship reaches age k . To see these two determinants, first, rewrite Eq. (15) to:

$$\Pi_k = \sum_{s=0}^{K-k} \left(\frac{\text{alive}_{k+s}^{ij}}{\text{alive}_k^{ij}} \times (\Theta_{k+s}^i)^\sigma \right) \times C$$

Expected profits rise, as remaining life expectancy, $\text{RLE}_k = \sum_{s=0}^{K-k} \frac{\text{alive}_{k+s}^{ij}}{\text{alive}_k^{ij}}$, and expected revenues, proportional to $(\Theta_k)^\sigma$, increase with relationship age. These two effects are illustrated in Figure A.5, with Panel A plotting remaining life expectancy and Panel B plotting expected revenues. According to the calibration, across the four countries, an old relationship has a remaining life expectancy that is 3 to 6 times larger than that of a new relationship. Expected sales also rise steeply with relationship age, being 17 to 40 times larger for old relationships across the four countries than for new relationships.

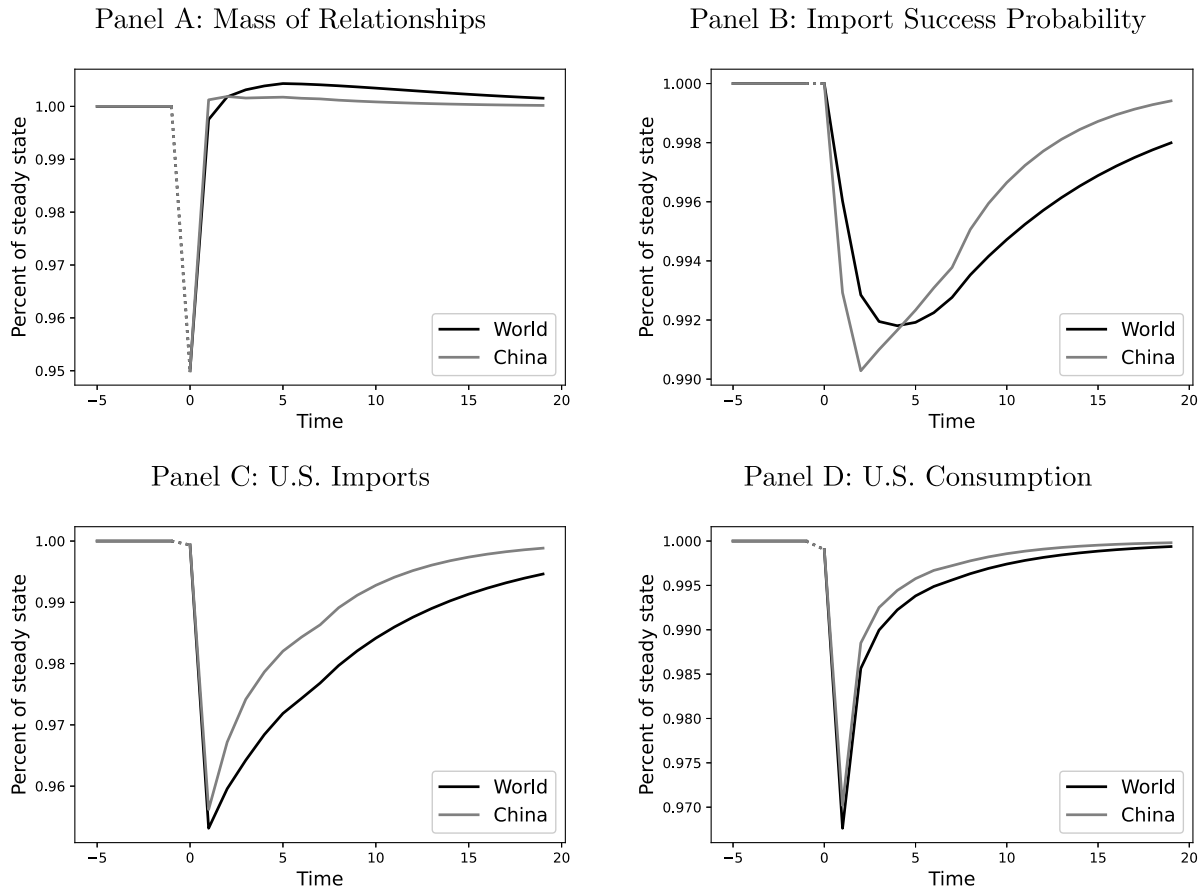


Fig. 9. Counterfactual: Destroying 5 percent of relationships. This figure shows results from the counterfactual where 5 percent of relationships are destroyed and entry in the initial period is 5 percent below its steady state level. For each line, we solve the asymmetric model between Canada (to proxy for the United States) and the other country/country group.

To further emphasize the heterogeneity across countries, we next illustrate how the expected profits from a relationship change with relationship age. Fig. 8 plots Π_k from Eq. (15) for two economies — China and Japan.⁴⁰ The starting point of the lines are given by Eq. (14) — the expected profits from a new relationship, Π_0 . New relationships have relatively low expected profits in both economies. However, in the long run, expected profits increase substantially and diverge notably.

4.3. Relationship disruptions and economic outcomes

We now present quantitative results from simulating the full model. The model is solved as outlined in Section 3.3, using our calibrated Θ_k^i parameters. We explore the importance of relationships for aggregate welfare by studying how the economy responds to an aggregate shock that destroys some fraction of pre-existing relationships. As the model does not feature endogenous relationship destruction, we focus on two experiments that directly shock the distributions of relationships in the two countries. That is, in the experiments, relationship distributions start from their steady-state values minus the assumed shock. First, to illustrate the dynamics, we study a symmetric shock that destroys an equal share of relationships across all ages. Second, we use observed relationship destruction rates during the U.S.-China trade war as an exogenous input into our counterfactual simulations.⁴¹ The exercises provide quantifications of how important relationships are for aggregate outcomes when the full dynamics of the economy are taken into account.

In the experiments, we let firms respond optimally, studying the transition path back to the steady state of the economy. In the period of the initial shock, we set entry exogenously either to its steady-state level or below, in line with the shock, effectively assuming that entry needs one period to respond to the shock. Without this assumption, entry would jump instantaneously when older relationships are destroyed, which would be an undesirable and unrealistic feature.

⁴⁰ Figure A.6 plots this object for all 10 sources.

⁴¹ Our model does not feature endogenous exit of relationships. Therefore, we use observed exit rates to inform the counterfactual. Augmenting the theory to allow for heterogeneous firms to enter or exit relationships is a fruitful area for further research.

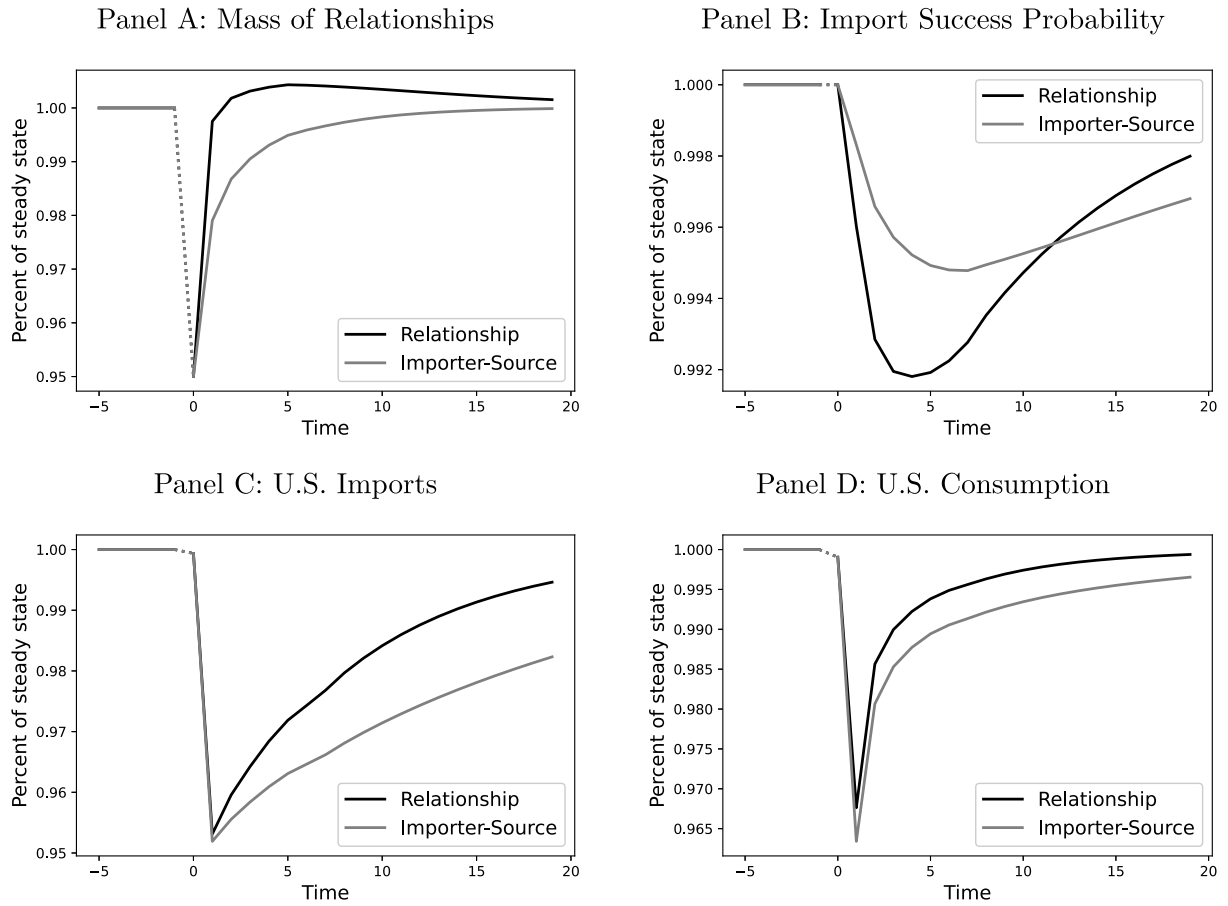


Fig. 10. Counterfactual: Destroying 5% of relationships, relationship vs. importer-source calibration. This figure shows results from the counterfactual where 5 percent of relationships are destroyed and entry in the initial period is 5 percent below its steady state level. For each line, we solve the asymmetric model between Canada (to proxy for the United States) and the World. The black line shows results for our baseline calibration at the relationship level. The gray line shows results for an alternative calibration using only data at the importer-source level.

We run each experiment separately, in each case solving an asymmetric two-country model between the United States and the other country (or the aggregate of all U.S. import sources, “World”).⁴² As we cannot estimate parameters for the United States with our data, we use the estimated θ_k from Canada for the United States. While the full set of results for all 10 major U.S. trading partners in the first experiment are shown in the appendix in Figure A.7 and Table B.8, we illustrate the main insights focusing on results for the World aggregate and for China in Figs. 9 through 11.

We first analyze the case in which 5 percent of relationships are destroyed symmetrically across all ages and entry in the initial period is set to 5 percent below its steady-state level. Results are shown in Fig. 9. Panel A shows the initial decline in the mass of relationships and the quick recovery through the entry of new firms. Panel B shows that the average import success probability ($\sum_{k=0}^K \text{alive}_k \theta_k$) falls initially, as new relationships have lower success probabilities than the average relationship that was destroyed by the shock. The average success probability then converges slowly back to its steady-state level, as the new relationships age. The reduced success probabilities in combination with the additional resources required to pay for the creation of new relationships cause U.S. imports and consumption (Panels C and D) to remain well below their steady state levels for many years. The cumulative reduction in U.S. imports from the shock is 40 percent of annual trade for the World and 23 percent of annual trade for China, with 62 percent and 46 percent of the cumulative import loss taking place more than 5 years after the initial shock for the World and China, respectively.

We also present a number of extensions and robustness checks for this experiment, including comparing the case of when the United States and World aggregate are assumed to have the same size versus the World aggregate being 5 times larger (Figure A.8), using a smoothed version of θ_k^i from age 7 onward (Figure A.9 and A.10), and calibrating an optimal σ for each country in addition

⁴² A multi-country version of the model would be an interesting extension: in a three-country model with asymmetric countries, there could be trade diversion effects. This is because relationships effectively reduce the marginal cost of trading. If old relationships between two countries are destroyed, this raises the marginal costs of trade between these two countries, leaving trade costs within other country-pairs unaffected.

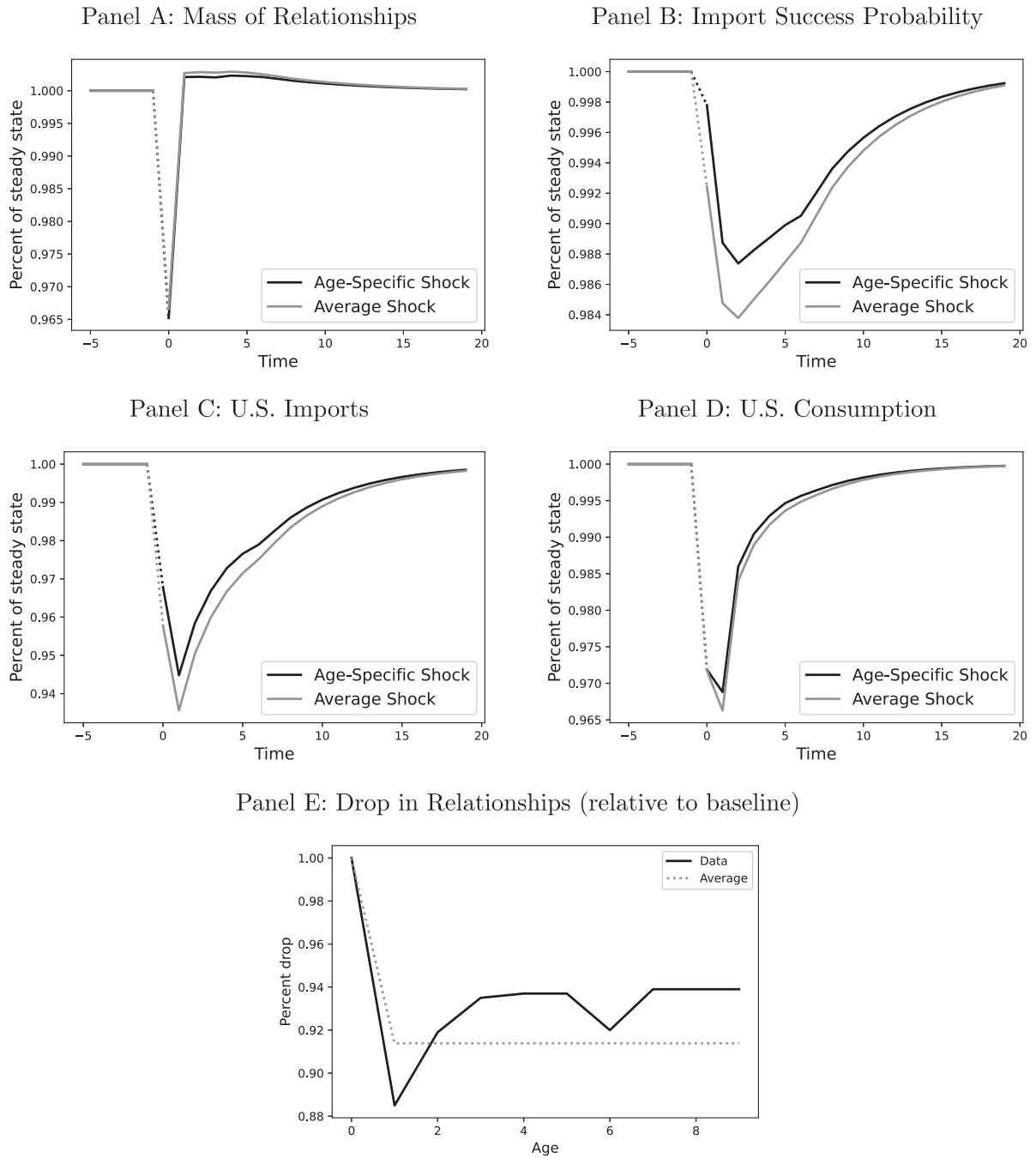


Fig. 11. Counterfactual: Trade war shock (imports from China). This figure shows results from the counterfactual where, in the initial period, relationships with Chinese suppliers are destroyed to match the decline in survival during the U.S.-China trade war, while keeping entry at its steady state level in the initial period. For each line, we solve the asymmetric model between Canada (to proxy for the United States) and China. The black line in Panels A through D shows results when relationship survival declines with age exactly as during the trade war. The gray line in Panels A through D shows results when relationship survival declines symmetrically across all ages. Panel E illustrates these shocks. The solid line shows the observed declines in survival rates, while the gray dotted line shows the symmetric shock.

to Θ_k^i (Figure A.11 and A.12). Varying σ affects the slope of the calibrated Θ_k as well as the CES aggregator, thereby impacting results on U.S. consumption quantitatively. Overall, however, results are in line with the main takeaways discussed above.

Next, returning to Fig. 9, we compare results for the World as a whole (black line) with results only based on Chinese suppliers (gray line). To understand the differences, first note that China has a larger share of young relationships and more trade in these

young relationships than the World as a whole. As a consequence, the estimated θ_k profile for China is well below the World aggregate estimates. Panel B of Fig. 9 shows that the import success probability for trade with China recovers notably more quickly than that for trade with the whole World, even though it drops by a bit more initially. In line with this pattern, the effects on U.S. imports and U.S. consumption (Panels C and D) reverse much more quickly for trade with China, reflecting the fact that relatively less relationship capital was destroyed since fewer relationships were old.

Before turning to our second experiment, we illustrate the importance of using relationship-level data to calibrate the model. Fig. 10 contrasts model results from a calibration to importer-supplier level data (our baseline) with those generated from a calibration to importer-source data. As shown in Table 1, older relationships trade much more and are much more prevalent when looking at more aggregate data. As a consequence, calibrating the model to importer-source data results in higher estimates of θ_k (see Figure A.13). In the counterfactuals, this generates a much smaller decline in the import success probability (Panel B) and a much slower recovery back to the steady-state levels of imports and consumption (Panels C and D). That is, when calibrating the model to the firm-source level, one misses important relationship dynamics that matter at the aggregate level. Because of this omission, the model underestimates the economy's ability to recover through the rebuilding of relationship capital.

In the second experiment the reduction in relationships in the initial period of the counterfactual exactly matches the observed relationship destruction from the U.S.-China trade war, while entry is fixed to its steady-state level on impact. Results are presented in Fig. 11. To compare the experiment to the first experiment and assess the relevance of the age-specific shocks, we also show the effect of a symmetric shock to relationship survival across all age groups that wipes out the same number of firms as the age-specific shock. The shocks are illustrated in Panel E. The solid line shows the actual drop in survival based on the percent changes calculated in Table 4. The dotted line shows the alternative symmetric shock applied to all relationships of age $k > 0$.

As before, the average success probability (Panel B) falls as a result of the relationship destruction. Since the actual trade war shock (solid line) is tilted towards younger relationships that have lower success probabilities, the decline is smaller than with a symmetric shock. Similarly, U.S. imports and consumption (Panels C and D) also drop by less with the age-specific shock than with the average shock.⁴³ Despite these differences, the recovery paths look surprisingly similar, indicating that even without the age-specific relationship survival information, the model does a good job at capturing the general dynamics.

To summarize the results, relationships represent a type of knowledge capital. Losing that capital is costly for an economy, implying a large drop in trade on impact and substantially lower levels of trade on the transition path, with corresponding losses in aggregate welfare. These effects can differ substantially across countries and depend on the country-specific patterns of relationship survival and sales over the life-cycle of relationships.

5. Conclusion

Relationships between importers and exporters are the core of international trade transactions. This paper presents new facts on these relationships and explores the value of trade relationships in two ways: first, it measures the value of relationships to firms by comparing expected profits across relationship ages within countries; second, it studies disruptions to trade relationships and their effects on aggregate efficiency, trade flows, and consumption. The impact on U.S. imports from a shock that destroys relationships is large and persistent. As we show, the quantitative results crucially depend on calibrating the model to the relationship-level data, as a calibration to more aggregate, firm-source level data implies a substantially slower recovery and larger trade losses in the model. We also document a notable heterogeneity among source countries in the relative importance of young and old relationships, which generates large differences in the estimated values of relationships across countries and in countries' dynamic responses to relationship destruction shocks.

This heterogeneity across source countries has several implications for the study of trade policy. First, our model suggests that the evolution of trade following Free Trade Agreements will depend on how trade is spread across relationships of different ages. At the same time, it implies that disruptions to relationships, such as from increased protectionism, may have significant trade and welfare effects. Indeed, although we show that the U.S.-China trade war tended to impact younger relationships more than older relationships, a further decoupling of the two economies would eventually impact older relationships to a larger extent as well, likely generating additional and more persistent negative effects on trade flows. More generally, our analysis suggests that accounting for the differences in the dynamics of relationship-level trade across sources will give more accurate predictions of the to correctly predict the short- to medium term dynamics of aggregate trade.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

<https://data.mendeley.com/datasets/yykpvjxsvj/1>.

⁴³ Note that the implications would be the reverse if the shock disproportionately destroyed older relationships, which have both higher delivery probabilities and more trade.

Appendix A-E. Supplemental figures, tables, and theory

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jinteco.2023.103842>.

References

- Aeberhardt, Romain, Buono, Ines, Fadinger, Harald, 2014. Learning, incomplete contracts and export dynamics: Theory and evidence from French firms. *Eur. Econ. Rev.* 68, 219–249.
- Albornoz, Facundo, Fanelli, Sebastián, Hallak, Juan Carlos, 2016. Survival in export markets. *J. Int. Econ.* 102, 262–281.
- Albornoz, Facundo, Pardo, Héctor F Calvo, Corcos, Gregory, Ornelas, Emanuel, 2012. Sequential exporting. *J. Int. Econ.* 88 (1), 17–31.
- Alessandria, George, Arkolakis, Costas, Ruhl, Kim J., 2021. Firm dynamics and trade. *Annu. Rev. Econ.* 13, 253–280.
- Alessandria, George, Choi, Horag, 2014. Establishment heterogeneity, exporter dynamics, and the effects of trade liberalization. *J. Int. Econ.* 94 (2), 207–223.
- Alessandria, George, Choi, Horag, Ruhl, Kim J., 2018. Trade adjustment dynamics and the welfare gains from trade. *J. Int. Econom.* 131, 103458.
- Alvarez, Fernando, Lucas, Robert E., 2007. General equilibrium analysis of the Eaton–Kortum model of international trade. *J. Monetary Econ.* 54 (6), 1726–1768.
- Antràs, Pol, Foley, C. Fritz, 2015. Poultry in motion: A study of international trade finance practices. *J. Polit. Econ.* 123 (4).
- Araujo, Luis F., Mion, Giordano, Ornelas, Emanuel, 2016. Institutions and export dynamics. *J. Int. Econ.* 98C, 2–20.
- Arkolakis, Costas, Papageorgiou, Theodore, Timoshenko, Olga A., 2018. Firm learning and growth. *Rev. Econ. Dyn.* 27, 146–168.
- Atkeson, Andrew, Burstein, Ariel Tomas, 2010. Innovation, firm dynamics, and international trade. *J. Polit. Econ.* 118 (3), 433–484.
- Bastos, Paulo, Dias, Daniel, Timoshenko, Olga, 2018. Learning, prices and firm dynamics. *Can. J. Econ.* 51 (4), 1257–1311.
- Benguria, Felipe, 2021. The matching and sorting of exporting and importing firms: Theory and evidence. *J. Int. Econ.* 131, 103430.
- Benguria, Felipe, Garcia-Marin, Alvaro, Schmidt-Eisenlohr, Tim, 2023. Trade Credit and Relationships. Working Paper.
- Bernard, Andrew B., Boler, Esther Ann, Massari, Renzo, Reyes, Jose-Daniel, Taglioni, Daria, 2017. Exporter dynamics, firm size and growth, and partial year effects. *Amer. Econ. Rev.* 107 (10), 3211–3228.
- Bernard, Andrew B., Jensen, J Bradford, Redding, Stephen J, Schott, Peter K, 2010. Wholesalers and retailers in US trade. *Amer. Econ. Rev.* 100 (2), 408–413.
- Bernard, Andrew B., Jensen, J. Bradford, Schott, Peter K., 2009. Importers, exporters and multinationals: A portrait of firms in the US that trade goods. In: *Producer Dynamics: New Evidence from Micro Data*. University of Chicago Press, pp. 513–552.
- Bernard, Andrew B., Moxnes, Andreas, Ulitveit-Moe, Karen Helene, 2018. Two-sided heterogeneity and trade. *Rev. Econ. Stat.* 100 (3), 424–439.
- Besedeš, Tibor, 2008. A search cost perspective on formation and duration of trade. *Rev. Int. Econom.* 16 (5), 835–849.
- Besedeš, Tibor, Prusa, Thomas J., 2006. Product differentiation and the duration of U.S. import trade. *J. Int. Econ.* 70 (2), 339–358.
- Blum, Bernardo S., Claro, Sebastian, Horstmann, Ignatius J., 2013. Occasional and perennial exporters. *J. Int. Econ.* 90 (1), 65–74.
- Carballo, Jerónimo, Ottaviano, Gianmarco IP, Martinus, Christian Volpe, 2018. The buyer margins of firms' exports. *J. Int. Econ.* 112, 33–49.
- Chaney, Thomas, 2014. The network structure of international trade. *Amer. Econ. Rev.* 104 (11), 3600–3634.
- Conte, Maddalena, Cotterlaz, Pierre, Mayer, Thierry, 2022. The CEPII Gravity Database. Working Paper 2022-05, CEPII.
- Drozd, Lukasz A., Nosal, Jaromir B., 2012. Understanding international prices: Customers as capital. *Amer. Econ. Rev.* 102 (1), 364–395.
- Eaton, Jonathan, Eslava, Marcela, Krizan, Cornell J, Kugler, Maurice, Tybout, James, 2014. A Search and Learning Model of Export Dynamics. NBER Working Paper.
- Eaton, Jonathan, Jinkins, David, Tybout, James, Xu, Daniel Yi, 2016a. Two-sided search in international markets. In: 2016 Meeting Papers. Society for Economic Dynamics.
- Eaton, Jonathan, Kortum, Samuel, 2002. Technology, geography, and trade. *Econometrica* 70 (5), 1741–1779.
- Eaton, Jonathan, Kortum, Samuel, Neiman, Brent, Romalis, John, 2016b. Trade and the global recession. *Am. Econ. Rev.* 106 (11), 3401–3438.
- Egan, Mary Lou, Mody, Ashoka, 1992. Buyer-seller links in export development. *World Dev.* 20 (3), 321–334.
- Fanelli, Sebastián, Hallak, Juan Carlos, 2018. Export Survival with Uncertainty and Experimentation. Mimeo.
- Garcia-Marin, Alvaro, Justel, Santiago, Schmidt-Eisenlohr, Tim, 2019. Trade Credit and Markups. CESifo Working Paper.
- Handley, Kyle, Kamal, Fariha, Monarch, Ryan, 2023. Supply Chain Adjustments to Tariff Shocks: Evidence from Firm Trade Linkages in the 2018–2019 US Trade War. Technical Report, National Bureau of Economic Research.
- Heise, Sebastian, 2015. Firm-to-Firm Relationships and Price Rigidity: Theory and Evidence. Mimeo.
- Heise, Sebastian, Pierce, Justin R, Schaur, Georg, Schott, Peter K, 2015. Trade Policy and the Structure of Supply Chains. Technical Report, Mimeo.
- Hillberry, Russell, Hummels, David, 2003. Intranational home bias: Some explanations. *Rev. Econ. Stat.* 85 (4), 1089–1092.
- Impullitti, Giammario, Irarrazabal, Alfonso A, Oromolla, Luca David, 2013. A theory of entry into and exit from export markets. *J. Int. Econ.* 90 (1), 75–90.
- Jovanovic, Boyan, 1982. Selection and the evolution of industry. *Econometrica* 50 (3), 649–670.
- Kamal, Fariha, Monarch, Ryan, 2018. Identifying foreign suppliers in US import data. *Rev. Int. Econ.* 26 (1), 117–139.
- Kamal, Fariha, Sundaram, Asha, 2016. Buyer–seller relationships in international trade: Do your neighbors matter? *J. Int. Econ.* 102, 128–140.
- Kaufmann, Daniel, Kraay, Aart, Mastruzzi, Massimo, 2010. The Worldwide Governance Indicators: Methodology and Analytical Issues. Policy Research Working Paper Series 4978, The World Bank.
- Krauthheim, Sebastian, 2012. Heterogeneous firms, exporter networks and the effect of distance on international trade. *J. Int. Econ.* 87 (1), 27–35.
- Krizan, C.J., 2012. The Nuts and Bolts of International Trade Transactions: A Data User's Primer. U.S. Census Center for Economic Studies, mimeo.
- Krolkowski, Pawel M., McCallum, Andrew H., 2021. Goods-market frictions and international trade. *J. Int. Econ.* 129, 103411.
- Krugman, Paul, 1980. Scale economies, product differentiation, and the pattern of trade. *Amer. Econ. Rev.* 70 (5), 950–959.
- Macchiavello, Rocco, Morjaria, Ameet, 2015. The value of relationships: evidence from a supply shock to Kenyan rose exports. *Amer. Econ. Rev.* 105 (9), 2911–2945.
- Martin, Julien, Mejean, Isabelle, Parenti, Mathieu, 2020. Relationship Stickiness, International Trade, and Economic Uncertainty. Discussion Paper DP15609, CEPR.
- McLaren, John, 1999. Supplier relations and the market context: A theory of handshakes. *J. Int. Econ.* 48 (1), 121–138.
- Monarch, Ryan, 2022. “It’s not you, it’s me”: Prices, quality, and switching in US-China trade relationships. *Rev. Econ. Stat.* 104 (5), 909–928.
- Pierce, Justin R., Schott, Peter K., 2016. The surprisingly swift decline of U.S. manufacturing employment. *Amer. Econ. Rev.* 106 (7), 1632–1662.
- Rauch, James E., 2001. Business and social networks in international trade. *J. Econ. Lit.* 1177–1203.
- Rauch, James E., Watson, Joel, 2003. Starting small in an unfamiliar environment. *Int. J. Ind. Organ.* 21 (7), 1021–1042.
- Rauch, James E., Watson, Joel, 2004. Network intermediaries in international trade. *J. Econ. Manag. Strategy* 13 (1), 69–93.
- Redding, Stephen J., Weinstein, David E., 2017. Aggregating From Micro to Macro Patterns of Trade. NBER Working Paper 24051.
- Reyes-Heroles, Ricardo, 2016. The Role of Trade Costs in the Surge of Trade Imbalances. Princeton University, mimeograph.
- Ruhl, Kim J., 2008. The International Elasticity Puzzle. Mimeo.
- Ruhl, Kim J., Willis, Jonathan L., 2017. New exporter dynamics. *Internat. Econom. Rev.* 58 (3), 703–726.
- Timoshenko, Olga A, 2015a. Learning versus sunk costs explanations of export persistence. *Eur. Econ. Rev.* 79, 113–128.
- Timoshenko, Olga A, 2015b. Product switching in a model of learning. *J. Int. Econ.* 95 (2), 233–249.